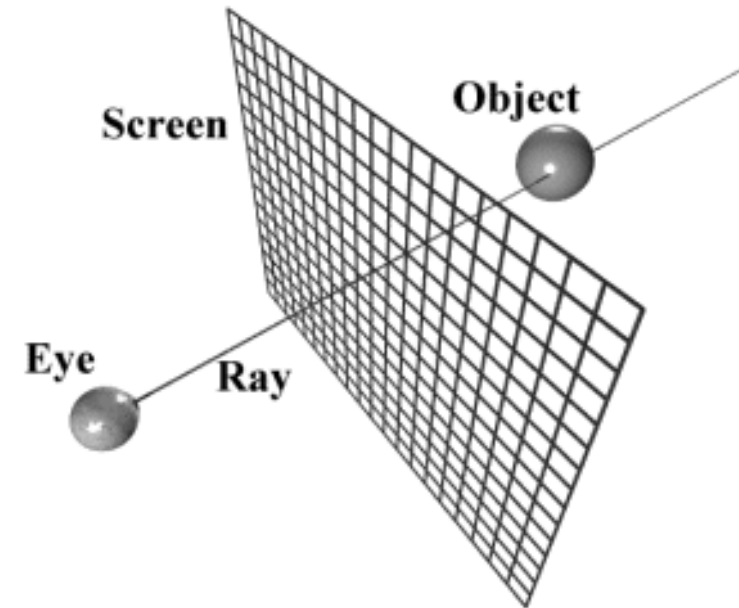
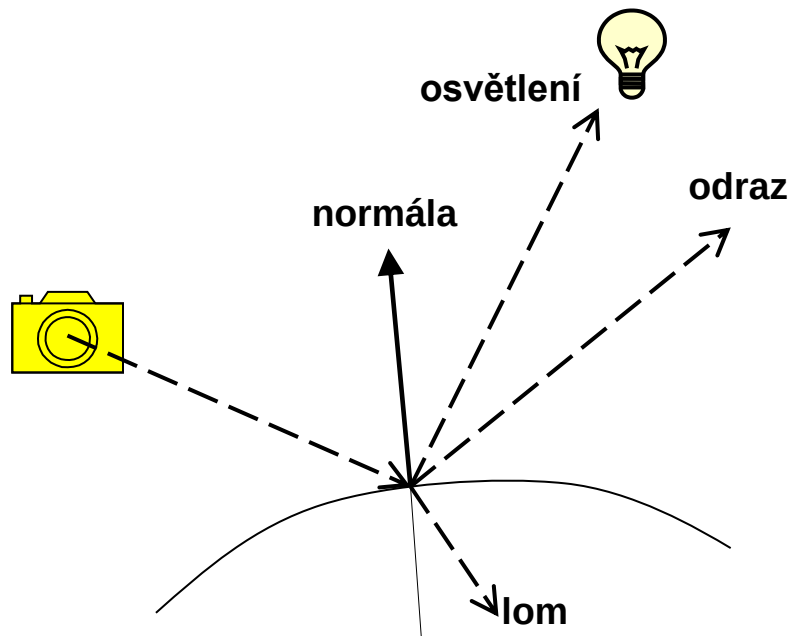


Distributed Ray Tracing Radiosity

prof. Ing. **Adam Herout**, PhD.

- Sledování paprsku – Ray tracing
 - Zpětné sledování
- Kamera, průmětna

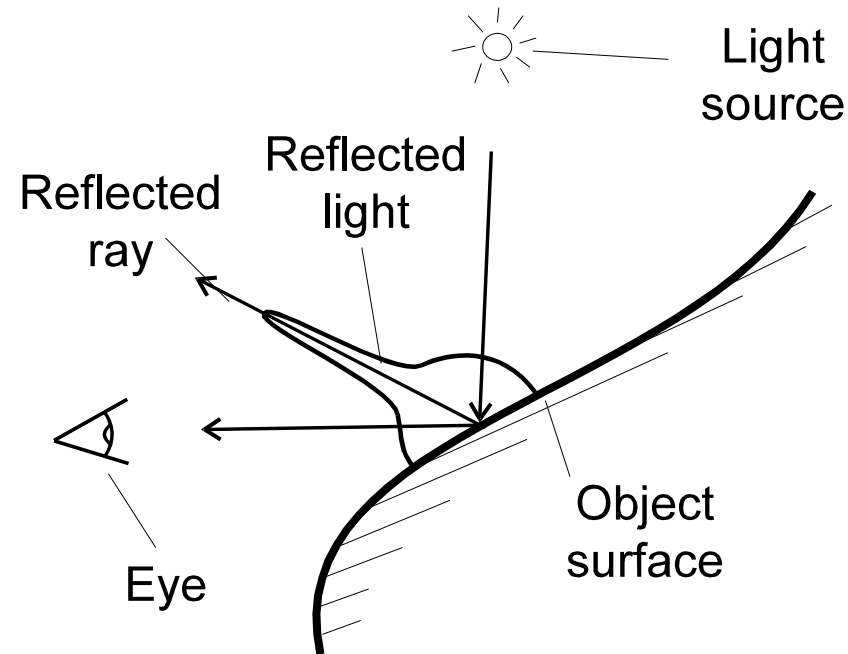


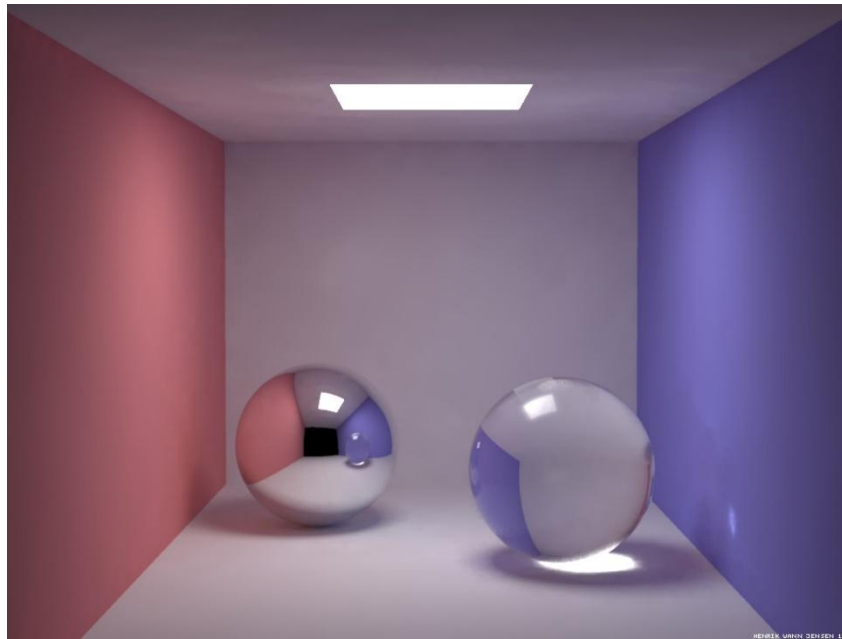
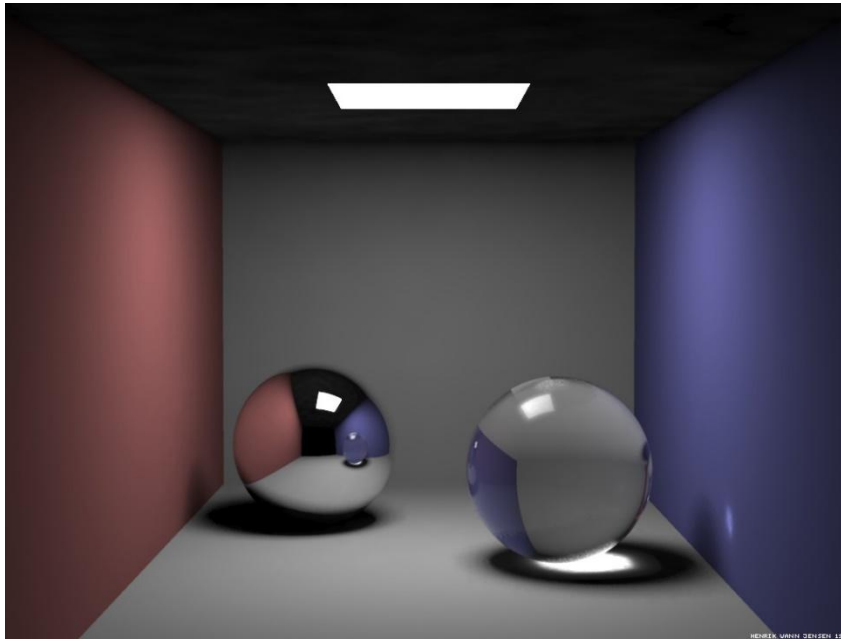
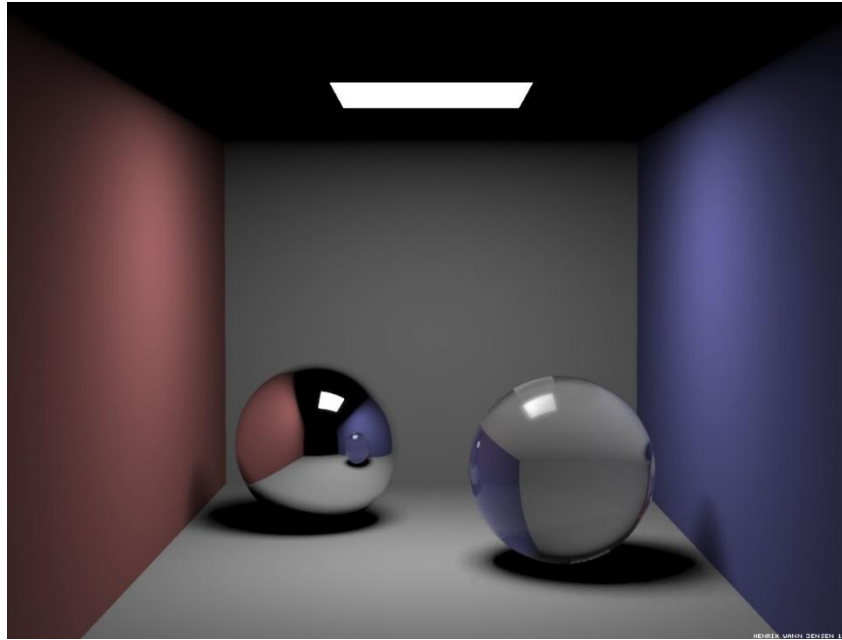
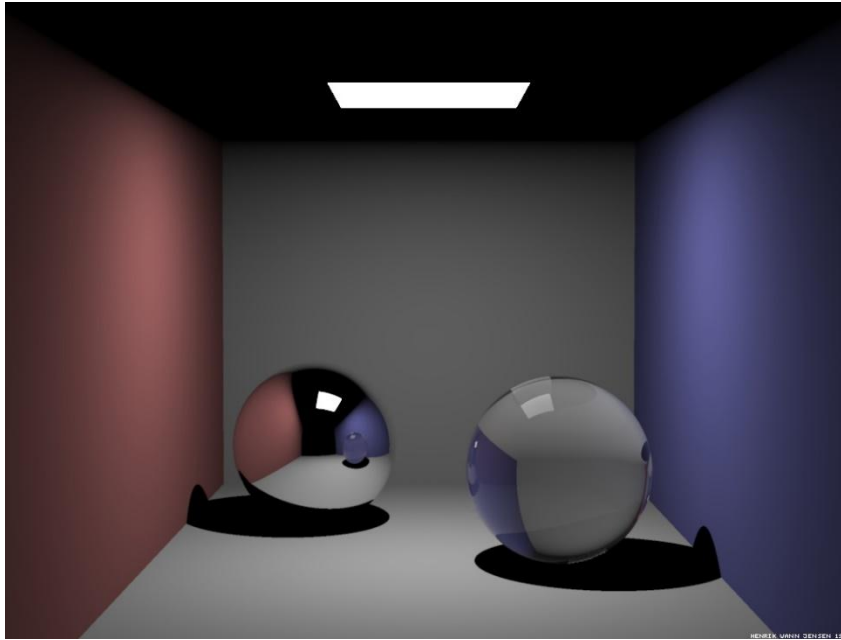
$$\mathbf{I}_d = \mathbf{I}_l k_d \cos \alpha = \mathbf{I}_l k_d (\mathbf{l} \cdot \mathbf{n})$$

$$\mathbf{I}_r = \mathbf{I}_l k_r \cos^n \beta = \mathbf{I}_l k_r (\mathbf{e} \cdot \mathbf{r})^n$$

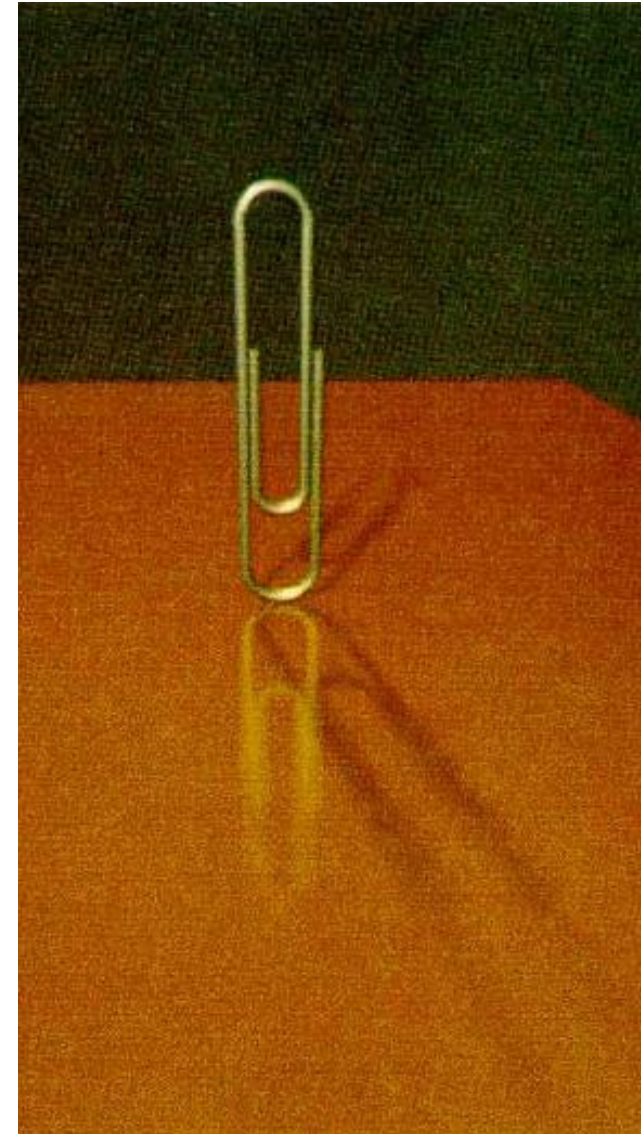
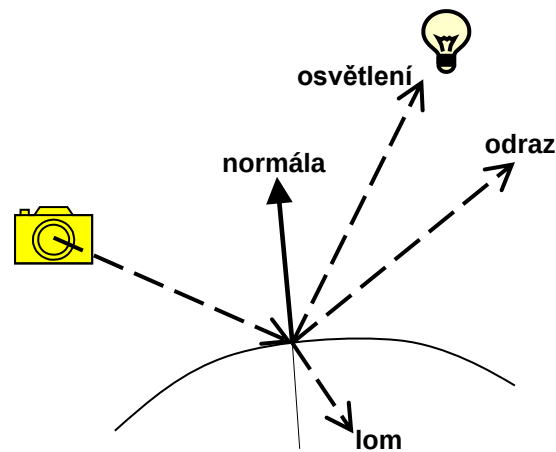
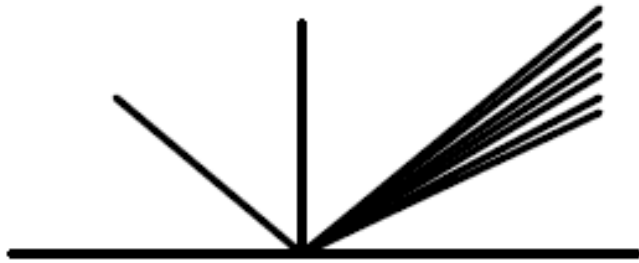
$$\mathbf{r} = \mathbf{l} - 2(\mathbf{l} \cdot \mathbf{n})\mathbf{n}$$

$$\mathbf{I} = \mathbf{I}_b + \mathbf{I}_d + \mathbf{I}_r$$

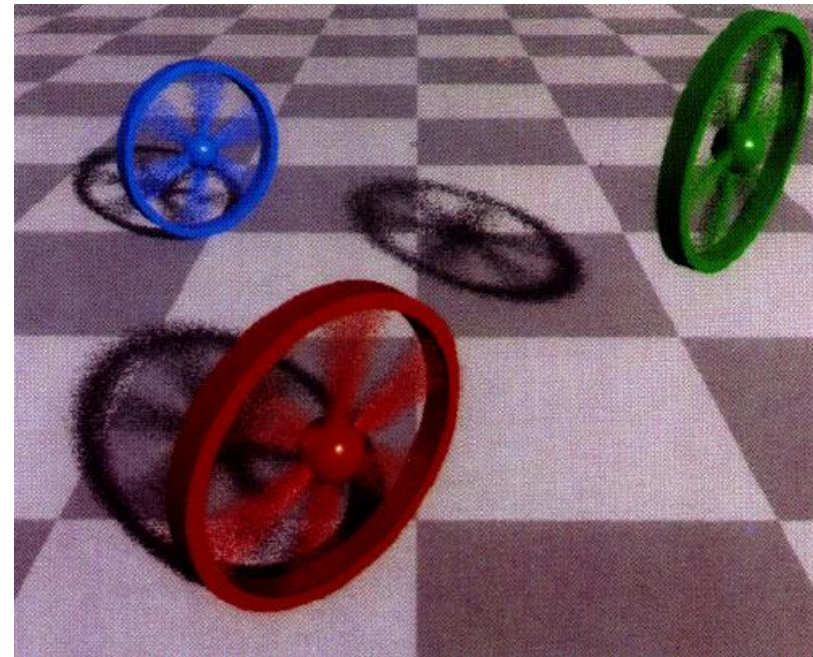




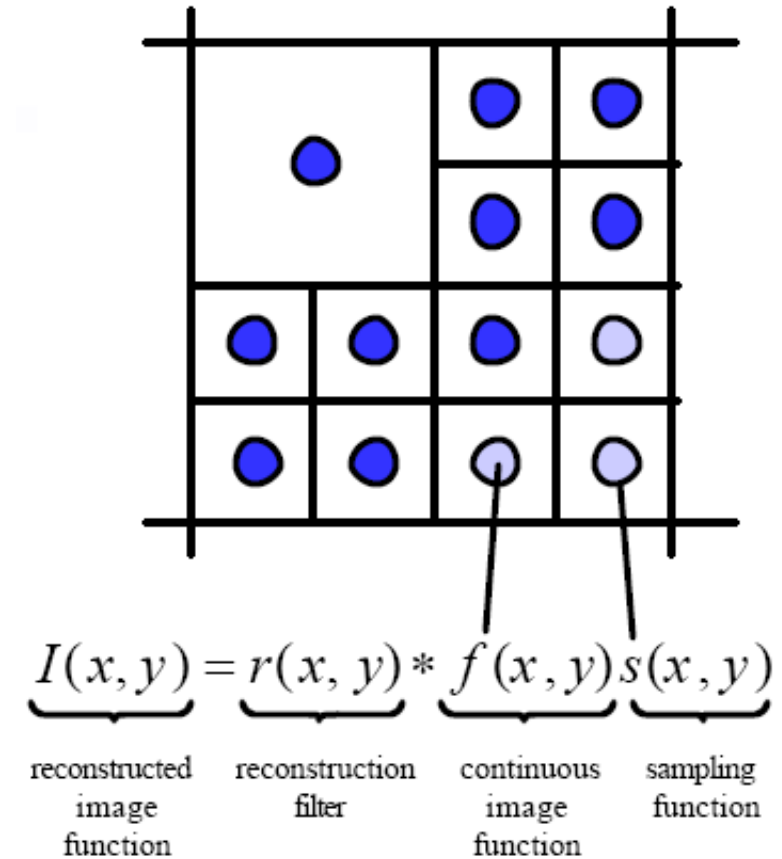
- Více odrazných paprsků
- Rekonstrukční filtr – největší příspěvek od těch vprostřed



- Paprsek nahrazen distribuční funkcí paprsků
- Funkce vzorkována dostatečným počtem paprsků
- Distribuční funkce
 - Paprsky z kamery (Eye rays)
 - Supersampling
 - Motion blur
 - Depth of field
 - Stínové paprsky (Shadow rays)
 - Měkké stíny
 - Odražené paprsky (Reflection rays)
 - Matné povrchy
 - Zalomené paprsky (Refraction rays)
 - Difúzní sklo



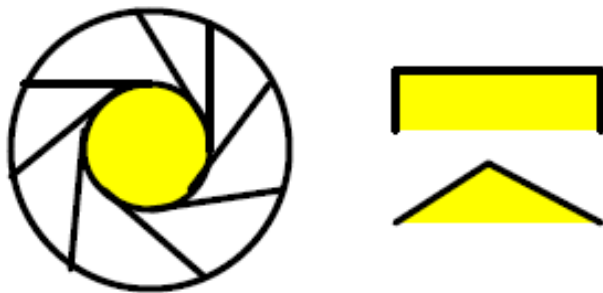
- Obrazová funkce $f(x,y)$
- Vzorkovací funkce $s(x,y)$
 - 1 když se vzorkuje v (x,y) , jinak 0
- Rekonstrukční filtr $r(x,y)$
 - Váhování podle plochy, polohy



$$I_{\text{pixel}} = \iint I(x, y) dx dy$$

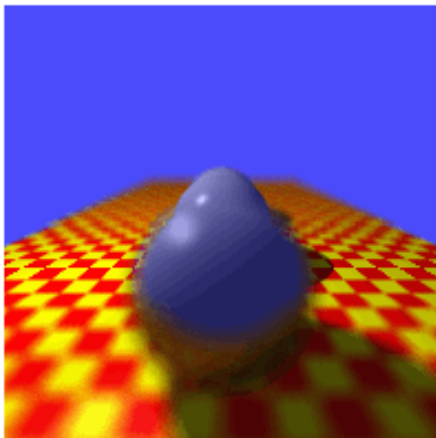
$$I_{\text{pixel}} = I\left(\frac{1}{2}, \frac{1}{2}\right)$$

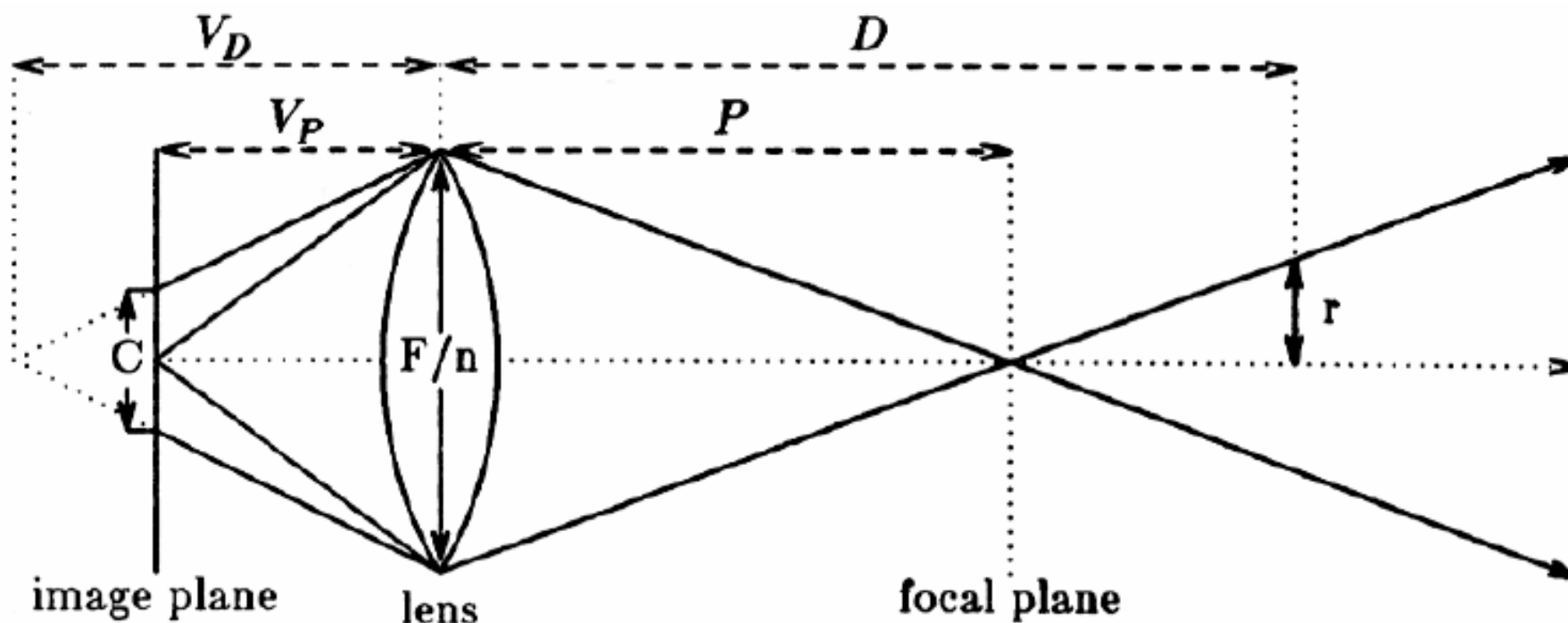
- Víc paprsků každým pixelem
- V různých časech
- Rychlost clony:
 - Rychlá: obdélníkový filtr
 - Pomalá: trojúhelníkový filtr



$$I(x, y, t) = \underbrace{r(t)}_{\text{shutter function}} * \underbrace{f(x, y, t)}_{\text{animated continuous image function}} \underbrace{s(t)}_{\text{temporal samples}}$$

- Lepší model čočky





F – focal length

n – aperture number

C – circle of confusion

$$V_P = FP / (P - F)$$

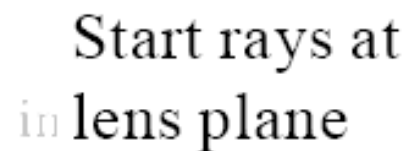
$$V_D = FD / (D - F)$$

$$C = (|V_D - V_P| / V_D) (F / n)$$

$$r = \frac{1}{2} (F / n) (D - P) / P$$

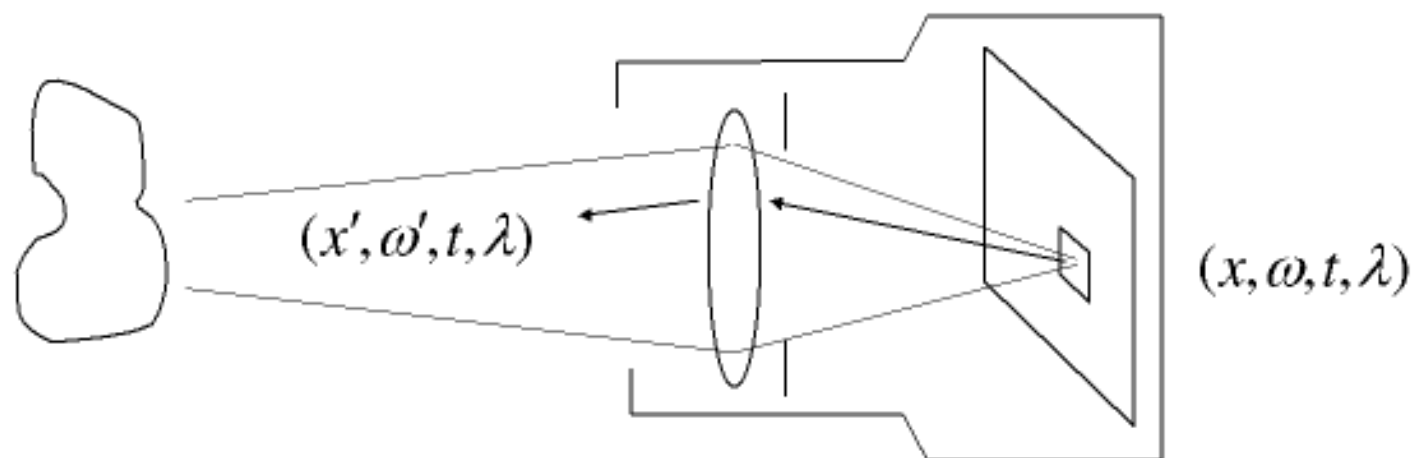
$$R = (-V_P / D) r$$

$$R = \frac{1}{2} C$$



- Pixel D uses ray BD
- Pixel E uses ray BE
- All rays emanate from B

- Pixel D uses rays AD, BD, CD
- Pixel E uses rays AE, BE, CE
- Rays emanate from lens plane



$$R = \iiint \iiint P(x, \lambda') S(x, \omega, t) L(T(x, \omega, \lambda), t, \lambda) d\vec{A} \bullet d\vec{\omega} dt d\lambda$$

Pixel response

$$P(x, \lambda)$$

Lens optics

$$(x', \omega') = T(x, \omega, \lambda)$$

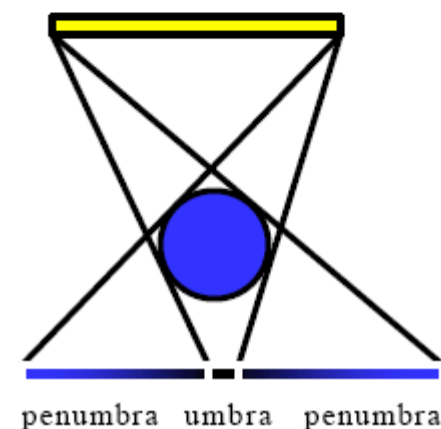
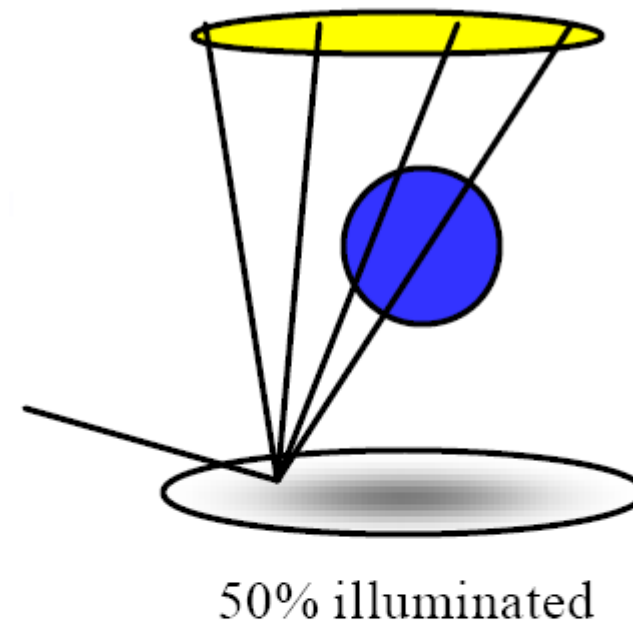
Shutter

$$S(x, \omega, t)$$

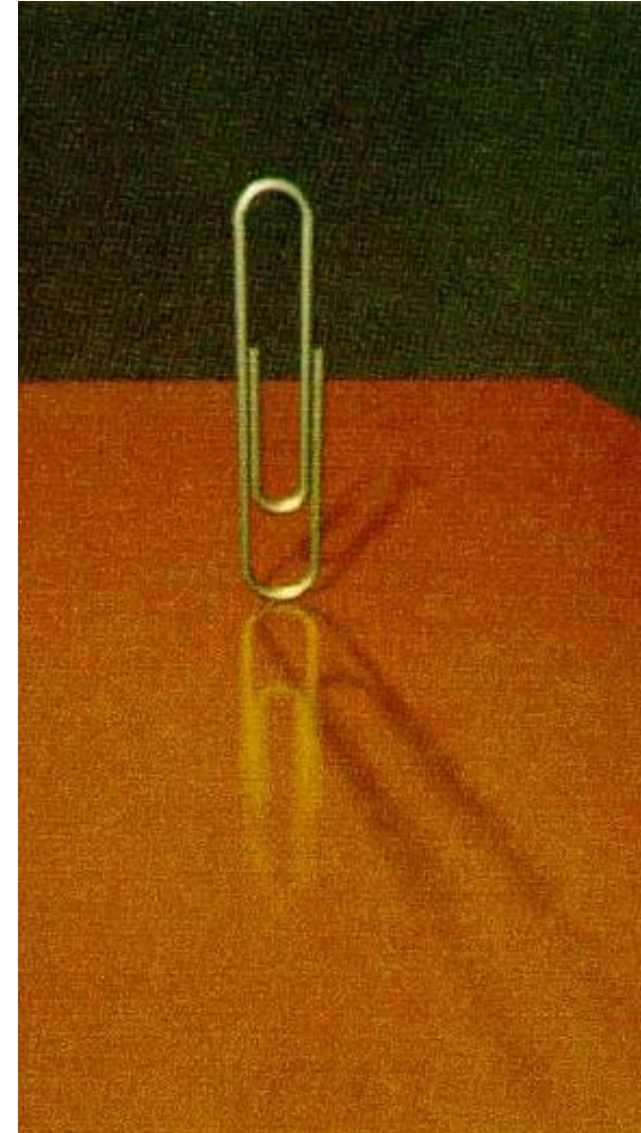
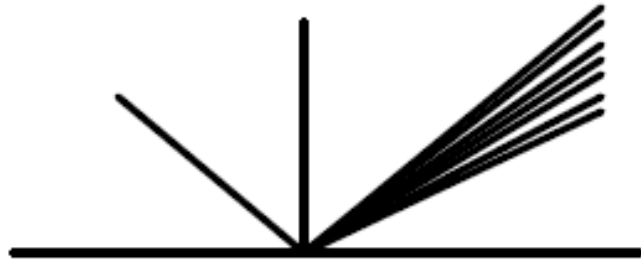
Scene radiance

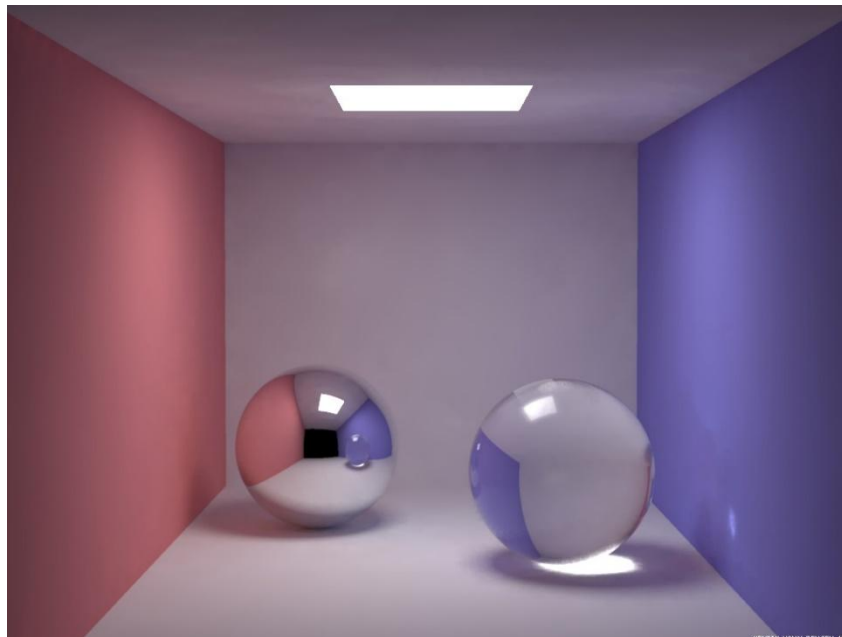
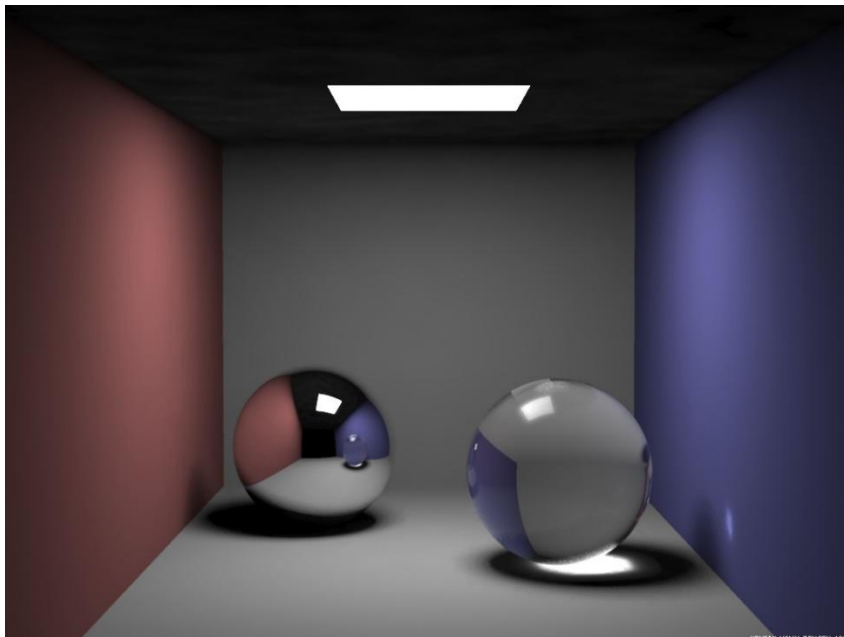
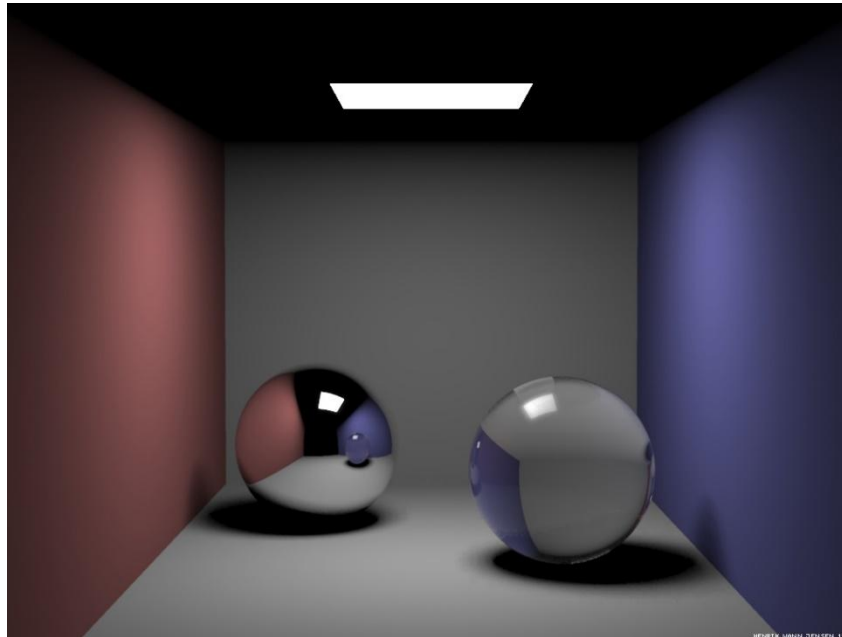
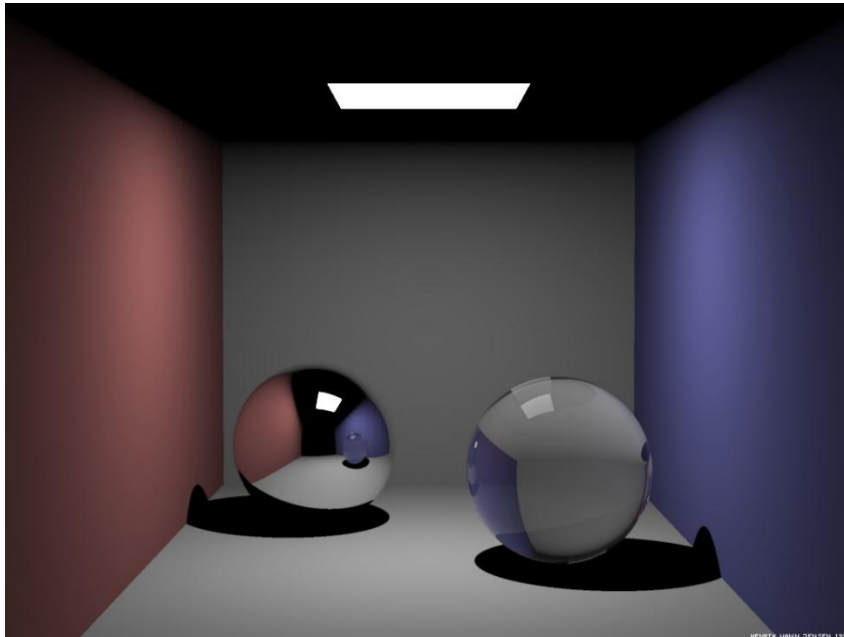
$$L(x, \omega, t, \lambda)$$

- Třeba použít plošné
- Stínové paprsky jsou vrhány z vyšetřovaného bodu do více míst ve zdroji světla
- Místa ve zdroji světla náhodně, aby se předešlo artefaktům



- Více odrazných paprsků
- Rekonstrukční filtr – největší příspěvek od těch vprostřed





Konstantní složka ambient



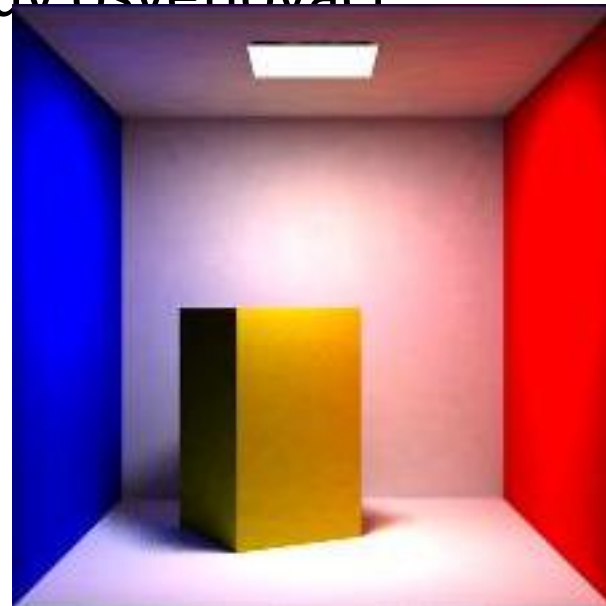
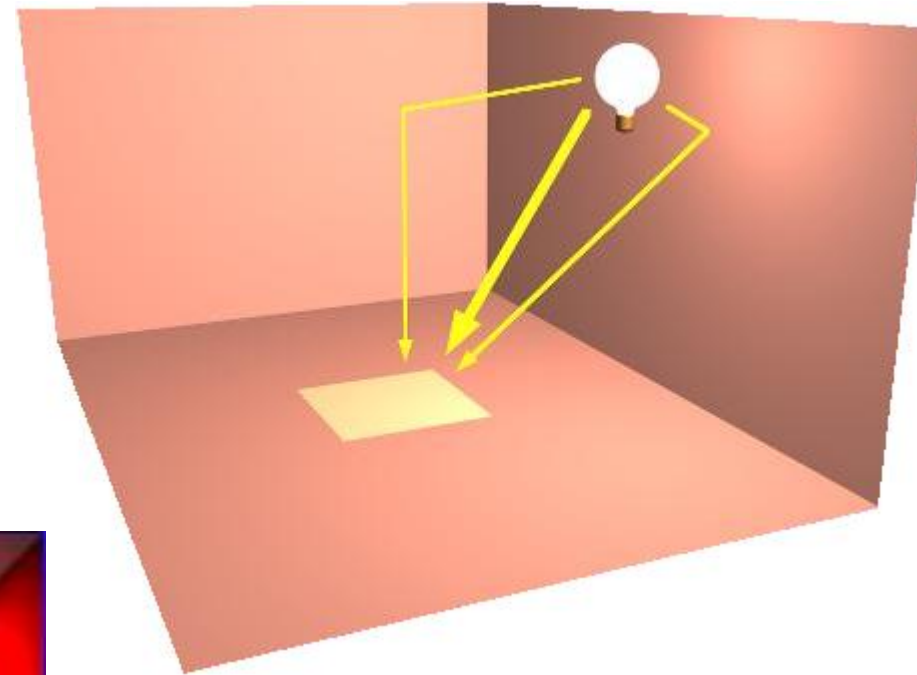
Bez složky ambient



Více průchodů difúzního osvětlení



- Každý difúzní povrch se stává virtuálním zdrojem světla
- Některé povrchy jsou opravdové zdroje světla
- Lambertův osvětlovací model



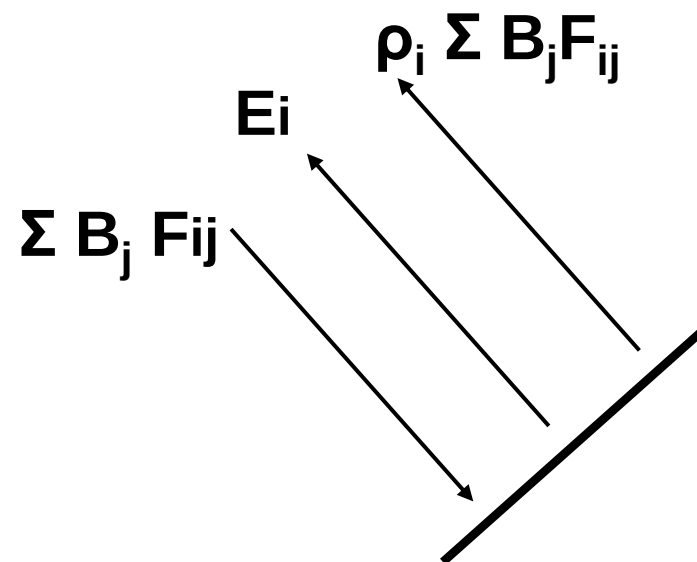
$$B_i = E_i + \rho_i \sum B_j F_{ij}$$

B_i – záření (radiosity) z povrchu i

E_i – vyzařování (emissivity) z povrchu i

ρ_i – odraznost (reflectivity) povrchu i

F_{ij} – Form Factor povrchu j oproti povrchu i



- Čistě geometrický vztah (nemá co dělat s materiály nebo umístěním kamery)

$$F_{dA_i dA_j} = \frac{\cos \phi_i \cos \phi_j}{\pi |r|^2}$$

dA_i, dA_j = differential area of surface i, j

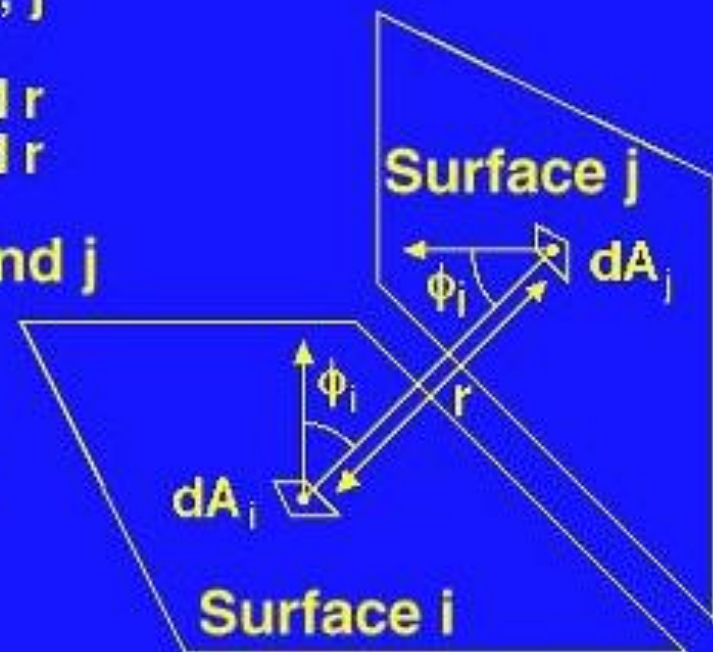
r = vector from dA_i to dA_j

ϕ_i = angle between Normal _{i} and r

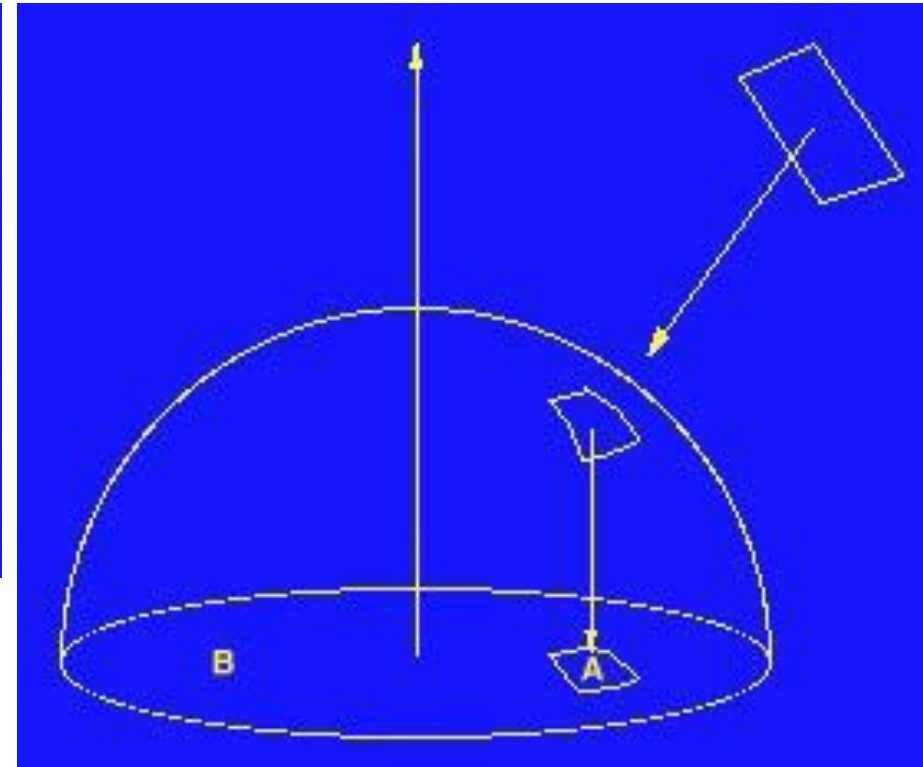
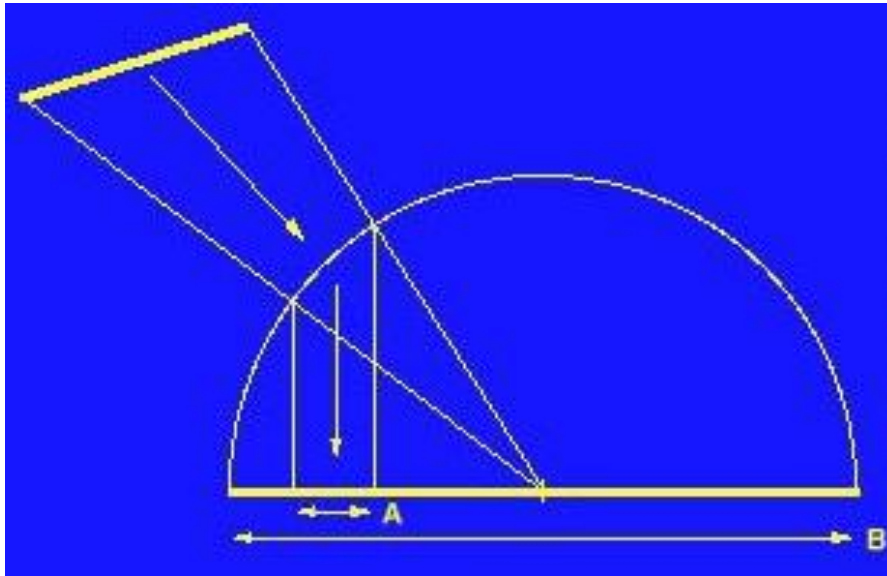
ϕ_j = angle between Normal _{j} and r

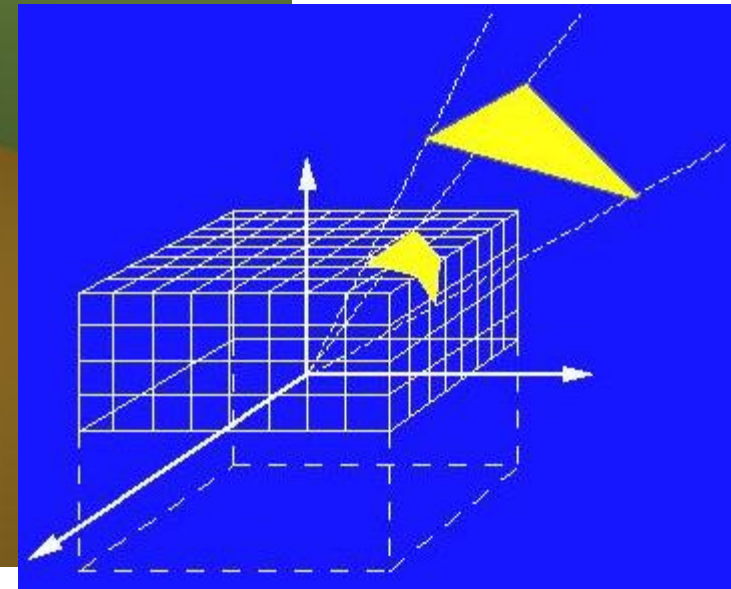
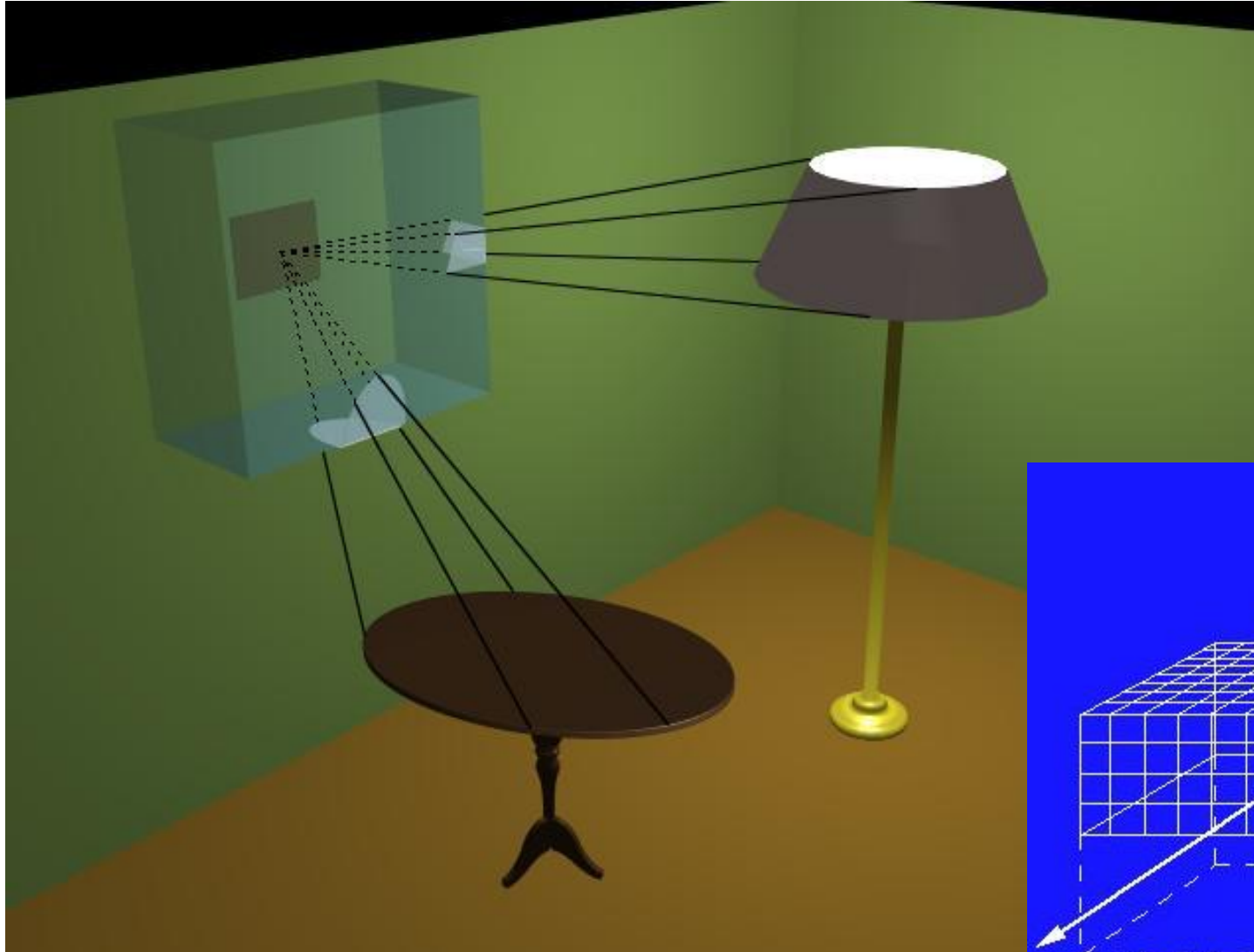
The overall form factor between i and j is found by integrating:

$$F_{ij} = \frac{1}{A_i} \int_{A_i} \int_{A_j} \frac{\cos \phi_i \cos \phi_j}{\pi |r|^2} dA_i dA_j$$



- Analogie k diferenciální definici f-f
- Umožňuje numerický výpočet



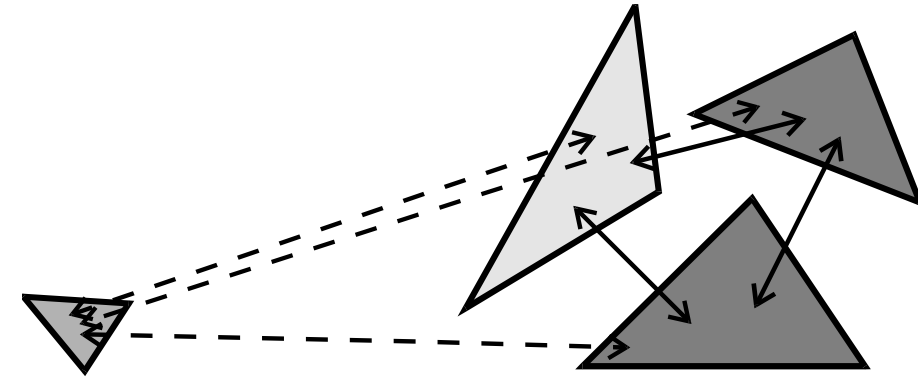


Radiosity on Graphics Hardware

Greg Coombe, Mark Harris,
Anselmo Lastra
UNC Chapel Hill

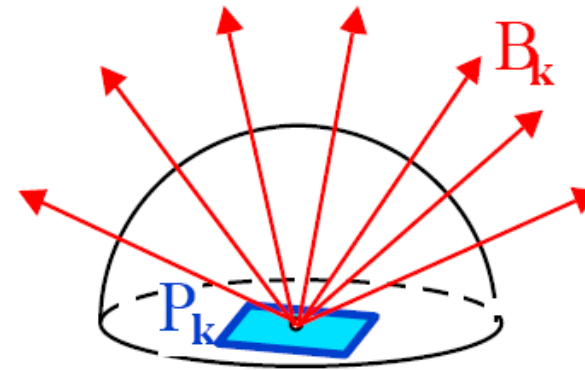
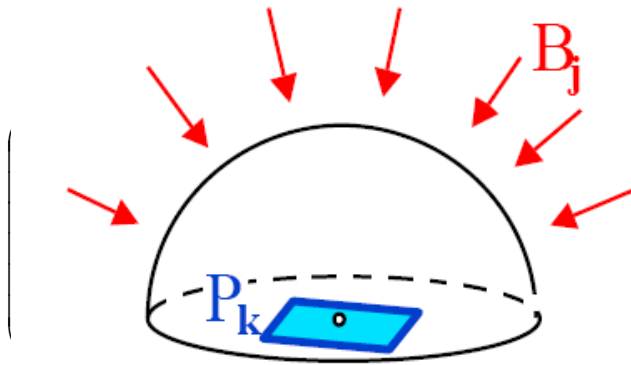
- Získat form-factory
- Přiřadit povrchům vlastní vyzařování
- Řešit rovnici
- Zobrazit pixelovou metodou (ray-tracing)
- Gauss-Seidelova

$$\begin{bmatrix} 1-\rho_1 f_{11} & -\rho_1 f_{12} & \cdots & -\rho_1 f_{1n} & \cdots & -\rho_1 f_{1n} \\ -\rho_2 f_{21} & 1-\rho_2 f_{22} & -\rho_2 f_{23} & & & \\ \vdots & -\rho_3 f_{32} & \ddots & & & \\ -\rho_i f_{i1} & & & 1-\rho_i f_{ii} & & \\ \vdots & & & & \ddots & \\ -\rho_n f_{n1} & & & & & 1-\rho_n f_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} l_1 \\ l_2 \\ \vdots \\ l_i \\ \vdots \\ l_n \end{bmatrix}$$





$$B_k^{i+1} = E_k + \rho_k \sum_{j \neq k} F_{kj} B_j^i$$



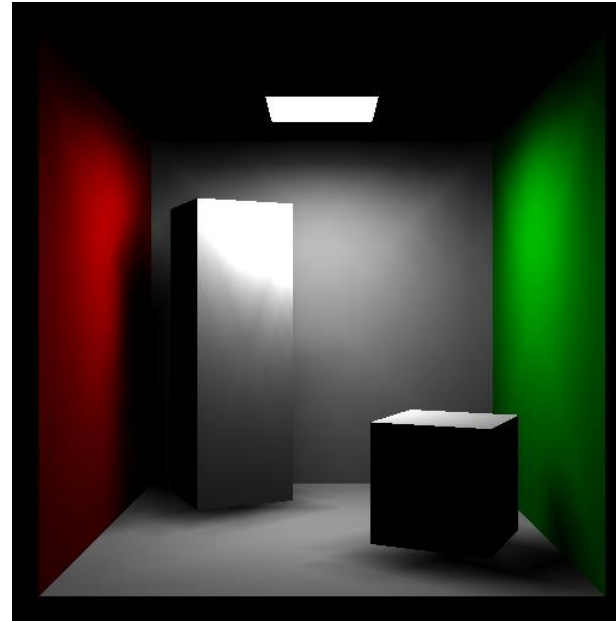
$$\begin{pmatrix} x \\ x \\ x \\ x \\ x \end{pmatrix} = \begin{pmatrix} x \\ x \\ x \\ x \\ x \end{pmatrix} + \begin{pmatrix} x \\ x \\ x \\ x \\ x \end{pmatrix} \cdot x$$

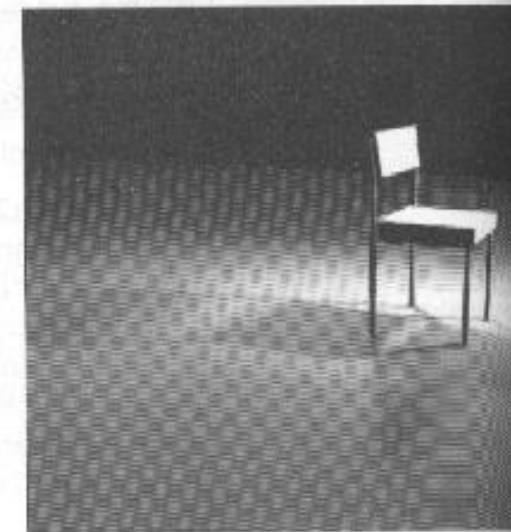
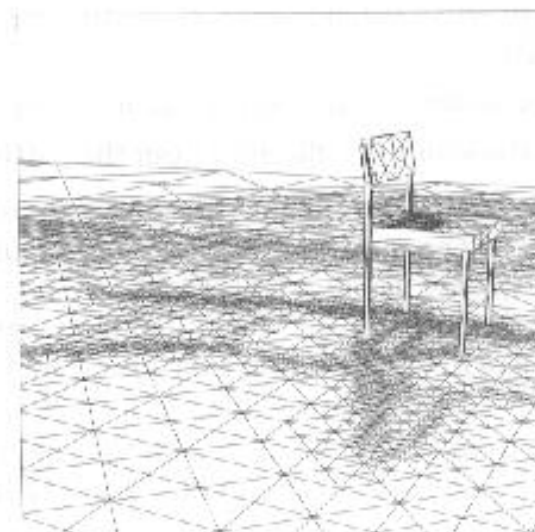
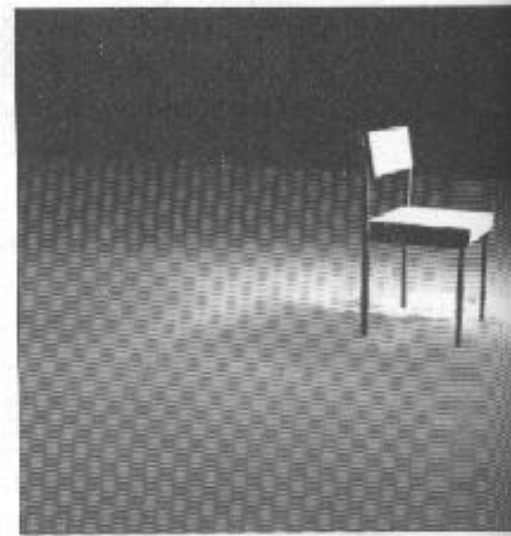
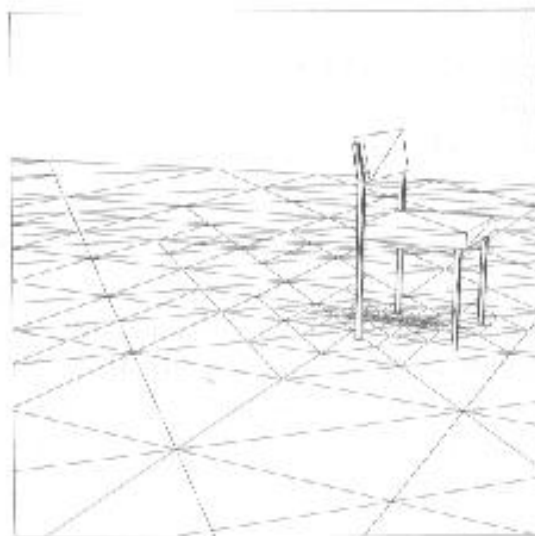
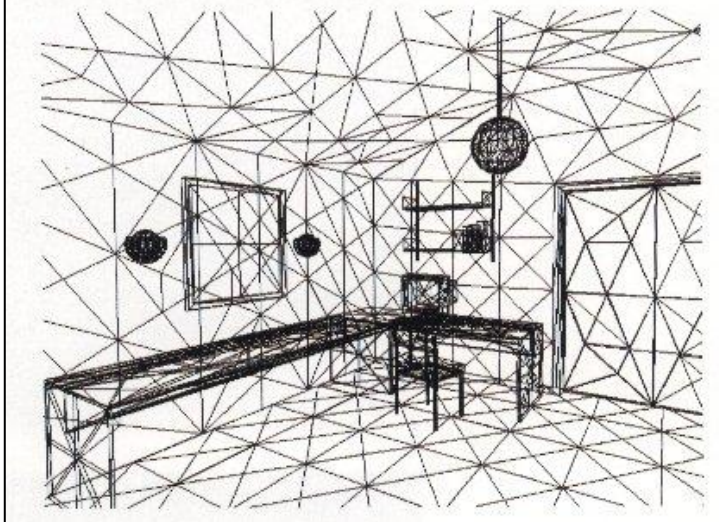
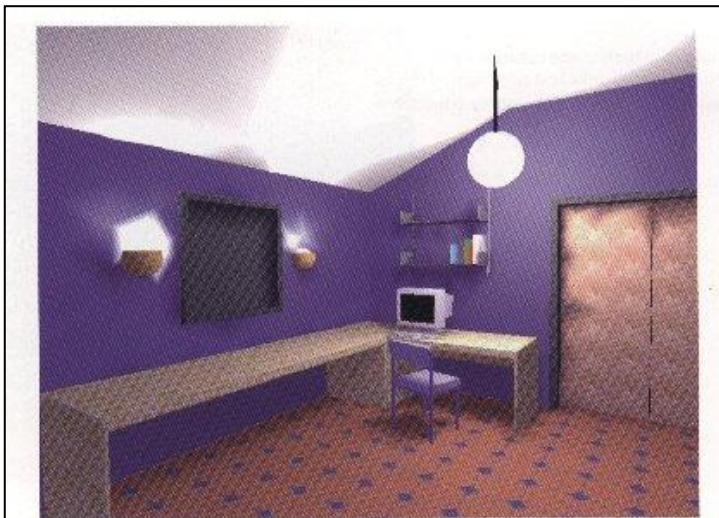
#1 Ze světelného zdroje

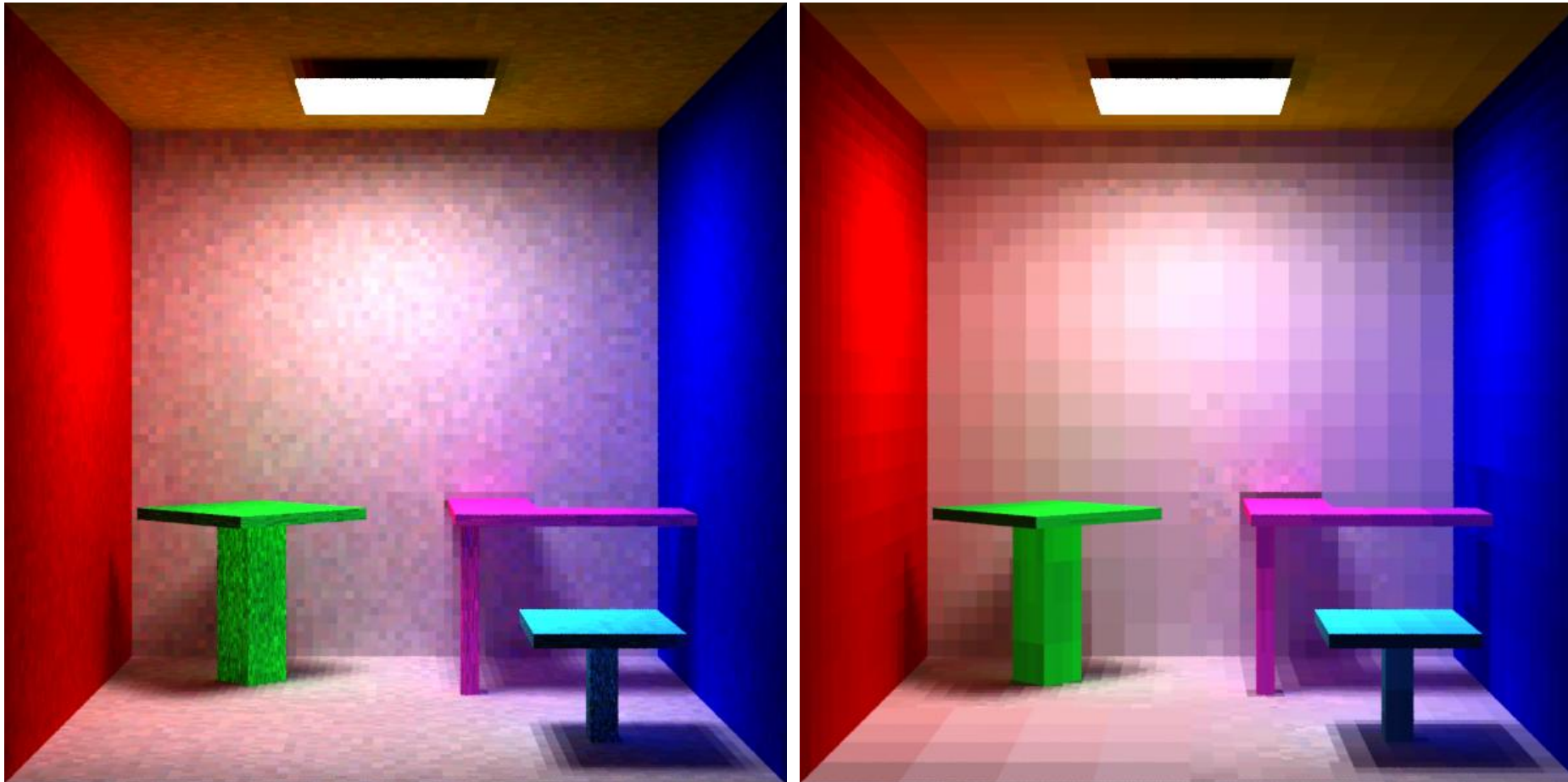
#4 Zelená zeď

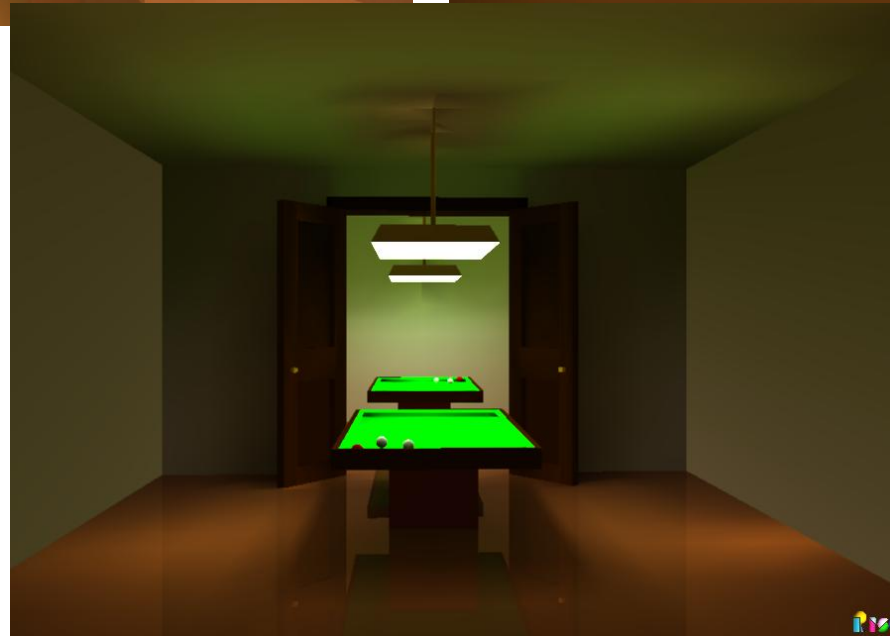
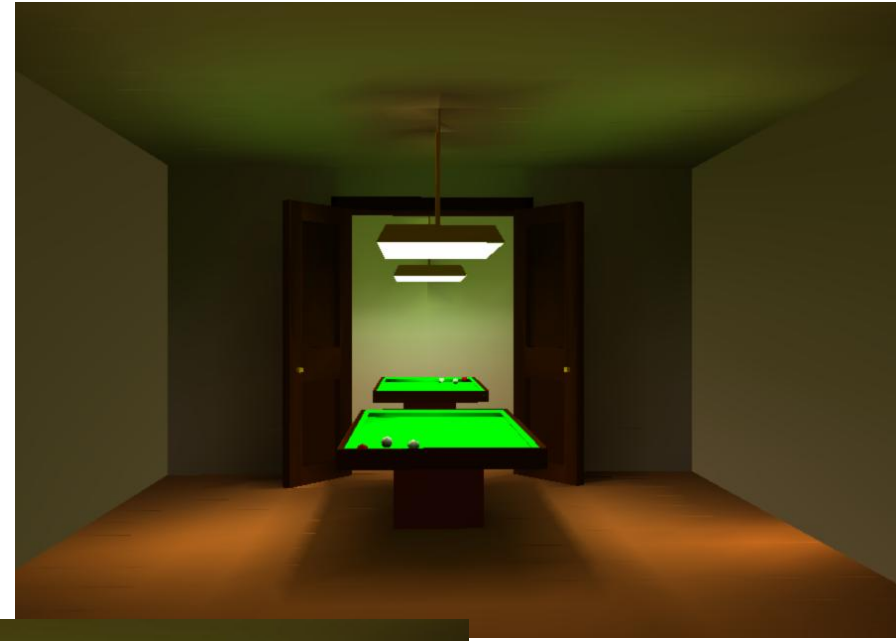
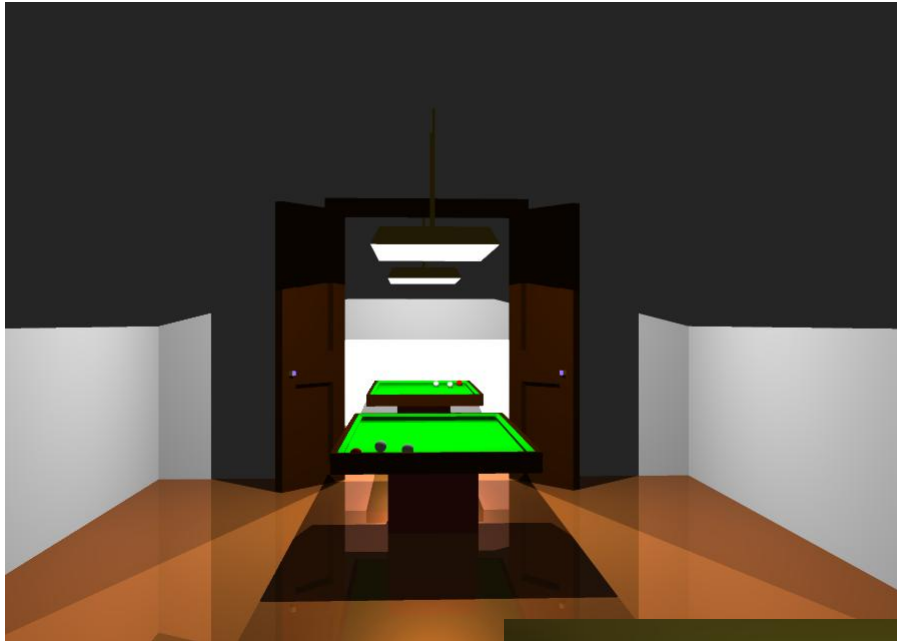
#7 Červená zeď

#24 Finální řešení





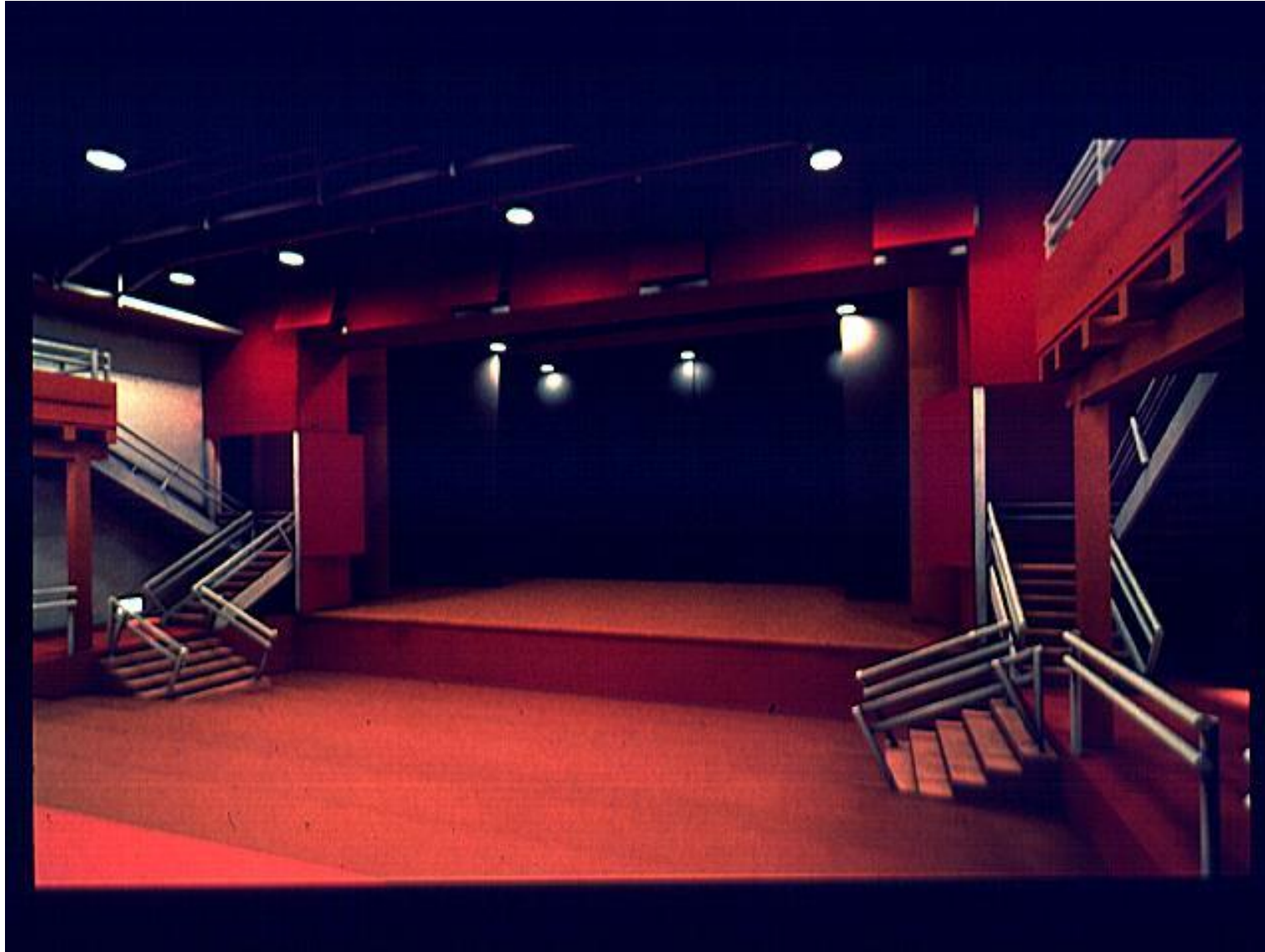










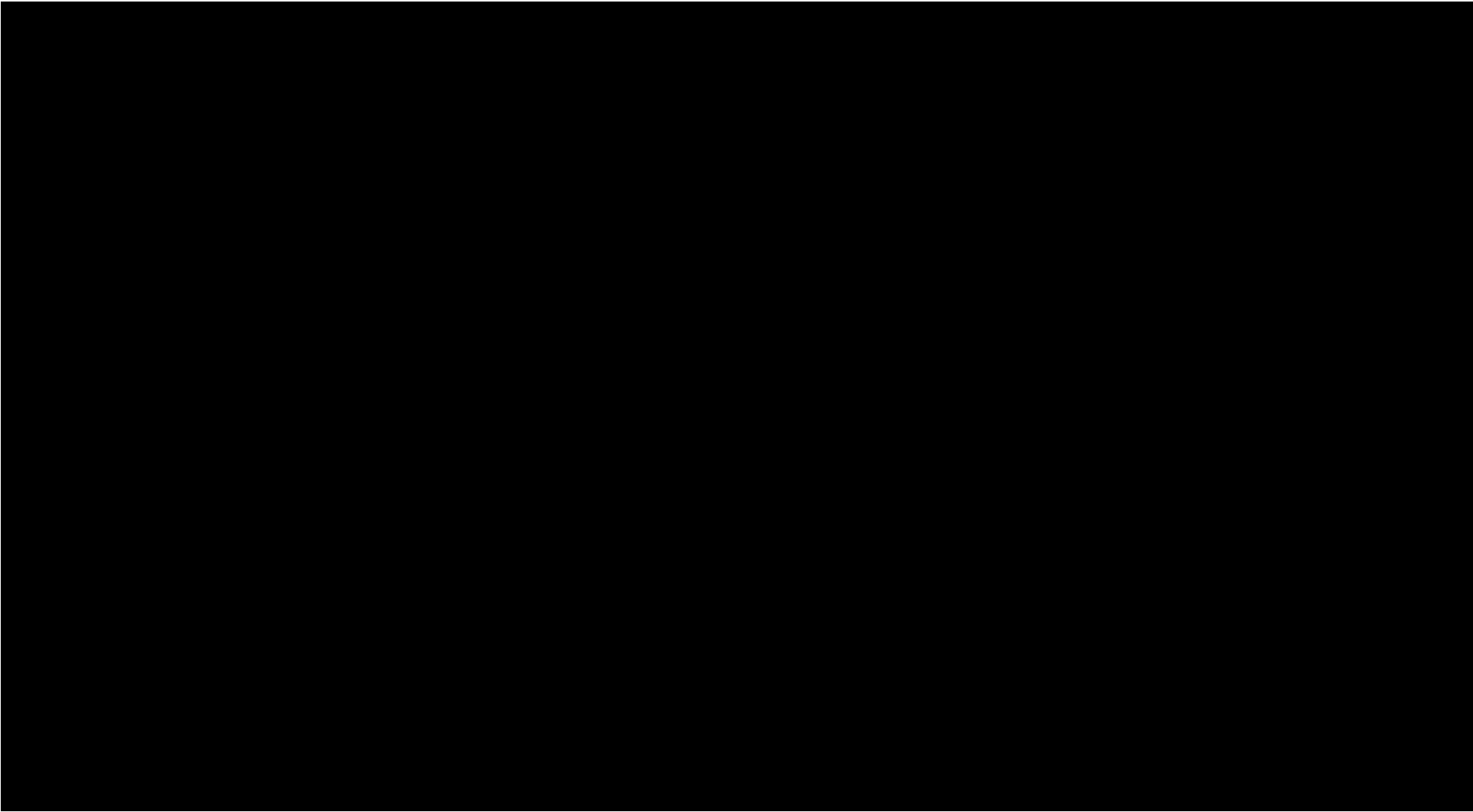


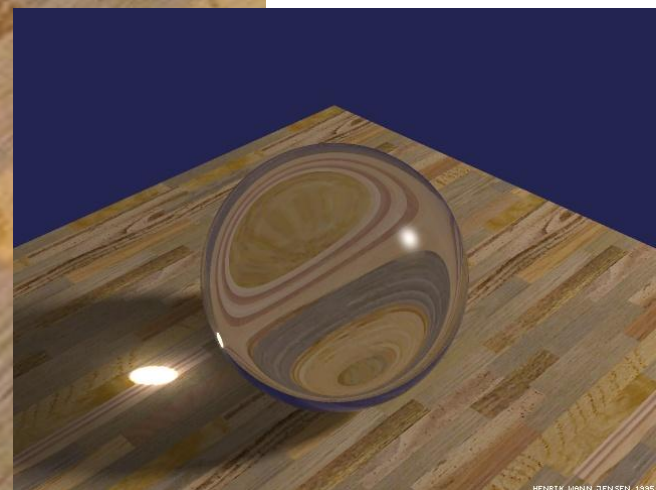
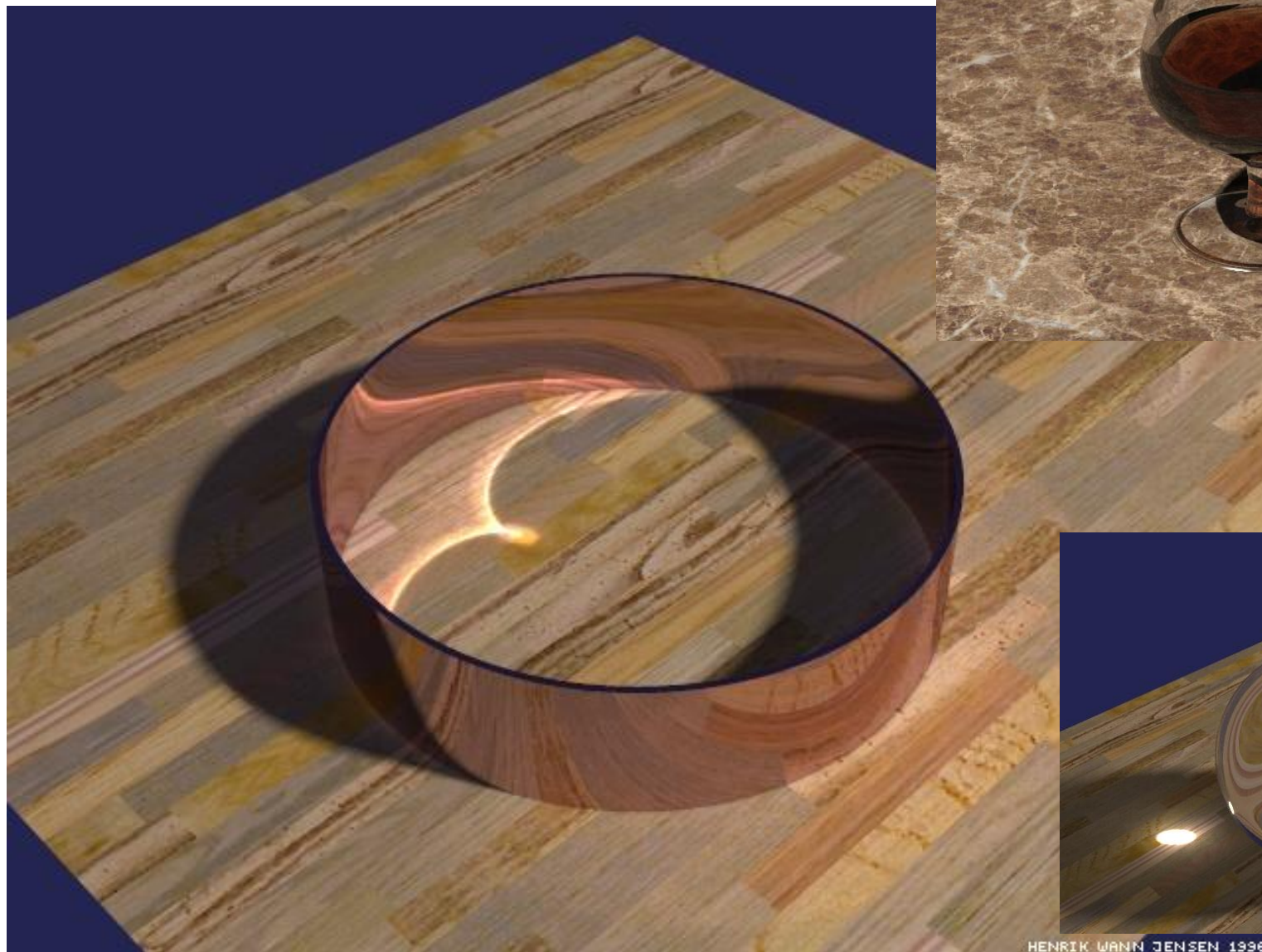












HENRIK WANN JENSEN 1996

HENRIK WANN JENSEN 1996



- Opravdové „Sledování paprsku“
- Fotony vystřelovány ze světla
- Zachytávány do textur
- Zobrazení „Zpětným sledováním paprsku“ nebo OpenGL

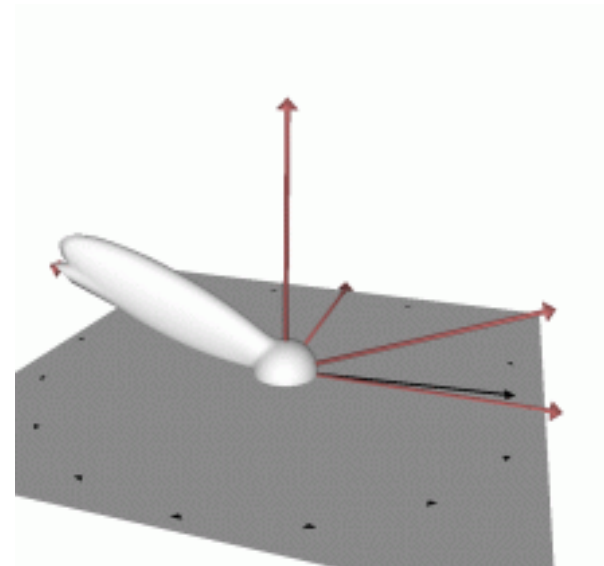
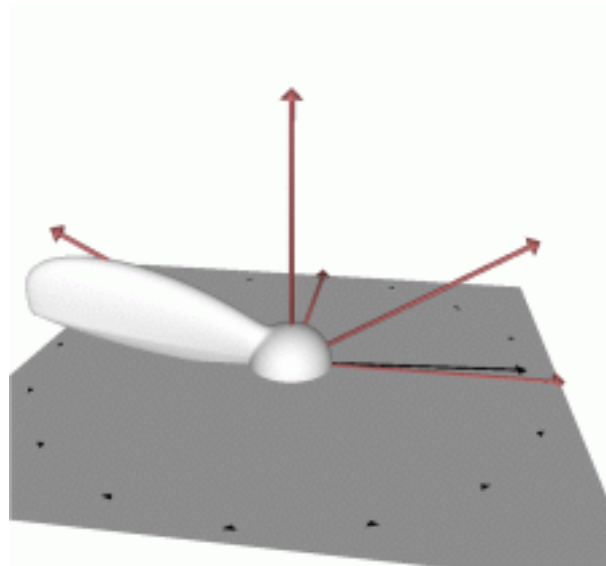
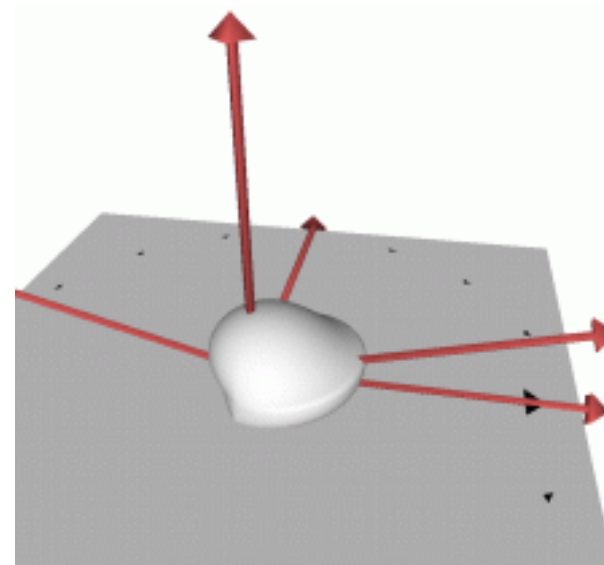
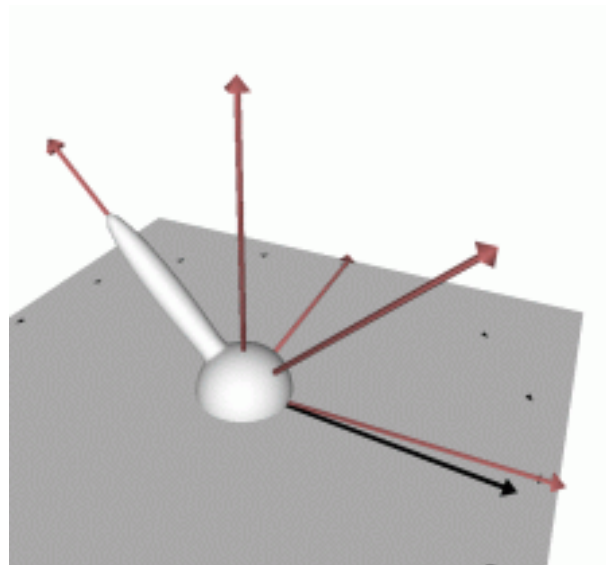
Realtime Caustics

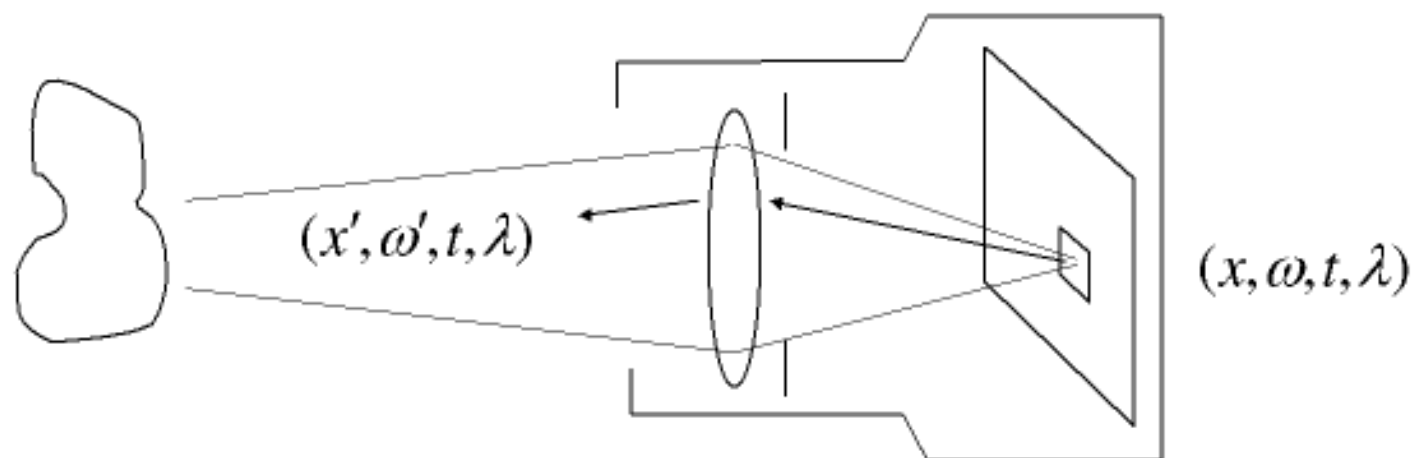
Using

Distributed Photon Mapping

Computer Graphics Group,
Saarland University, Germany

<http://graphics.cs.uni-sb.de>





$$R = \iiint \iiint P(x, \lambda') S(x, \omega, t) L(T(x, \omega, \lambda), t, \lambda) d\vec{A} \bullet d\vec{\omega} dt d\lambda$$

Pixel response

$$P(x, \lambda)$$

Lens optics

$$(x', \omega') = T(x, \omega, \lambda)$$

Shutter

$$S(x, \omega, t)$$

Scene radiance

$$L(x, \omega, t, \lambda)$$

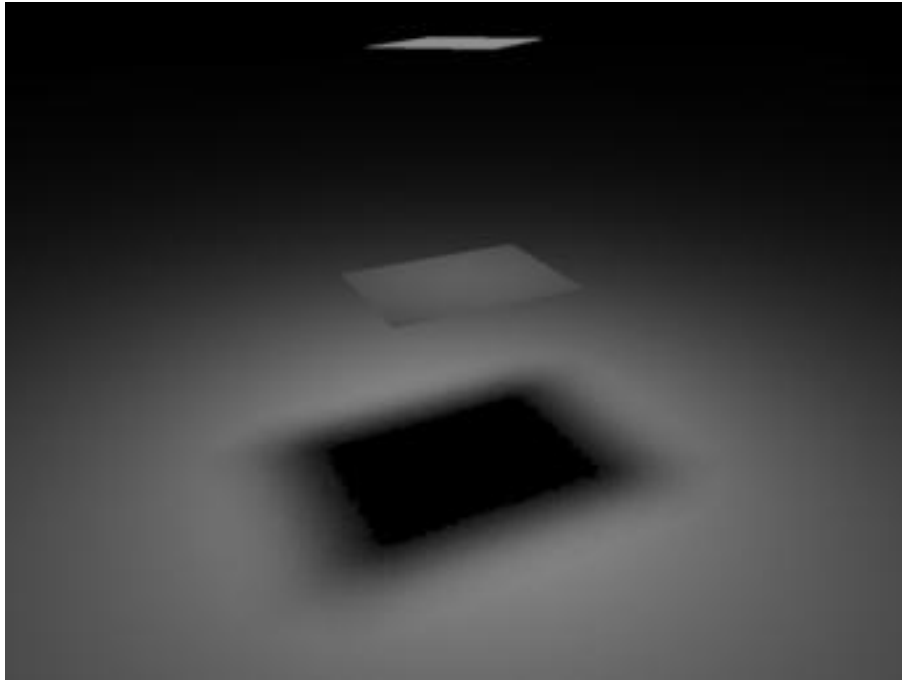


- Integrovat přes všechny cesty

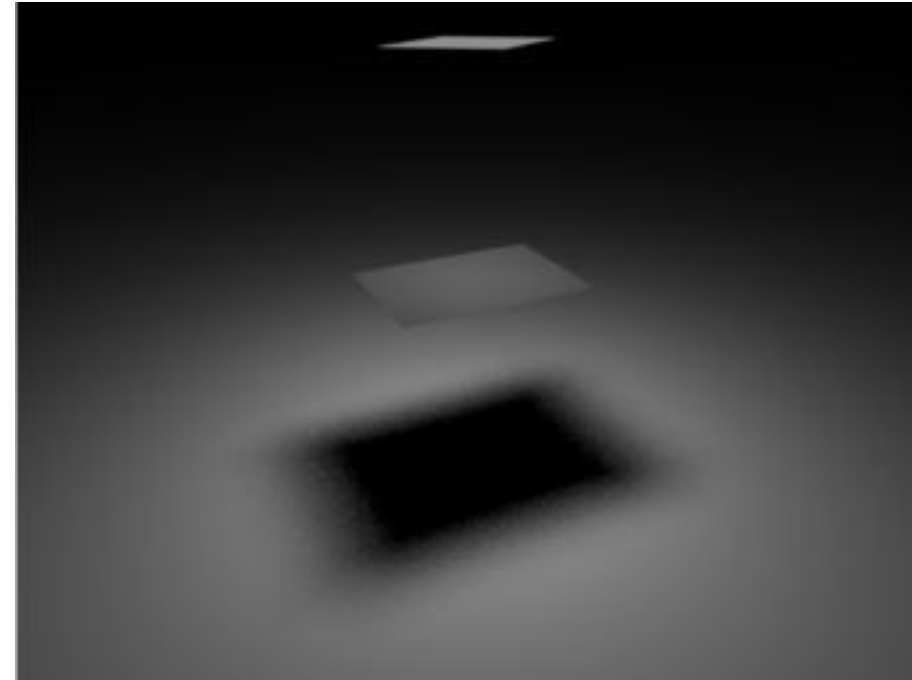
$$L(x_{k-1}, x_k) \\ = \sum_{k=1}^{\infty} \int_{M^2} \cdots \int_{M^2} L_S(x_0, \cdots, x_{k-2}, x_{k-1}, x_k) dA(x_0) \cdots dA(x_{k-2})$$

- Jak navzorkovat prostor všech cest?
- Dobré estimátory?

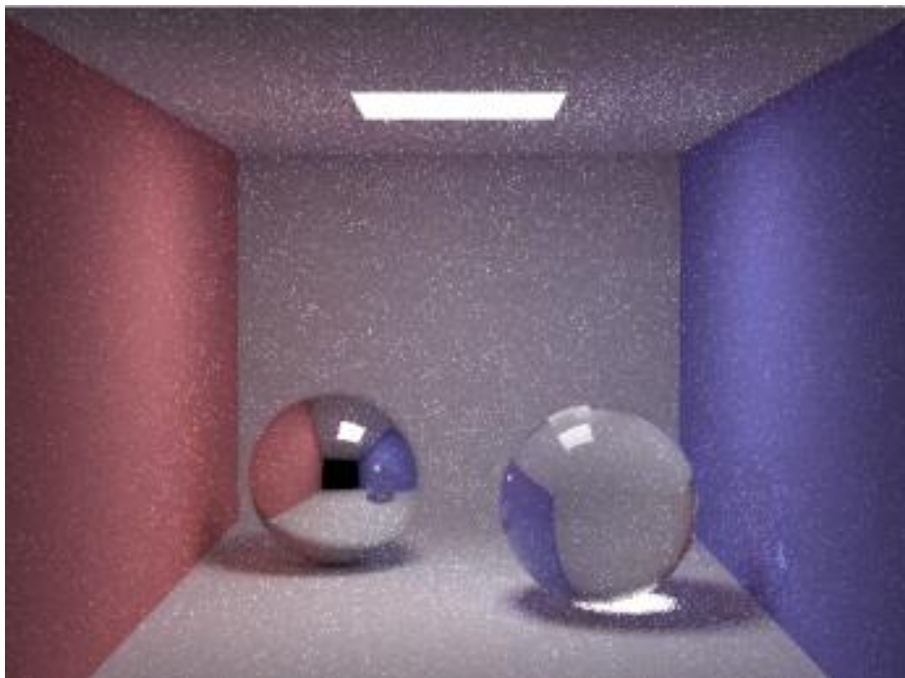
1. Zvolit paprsek (x,y, u,v, t)
 - Weight = 1
2. Sledovat paprsek
3. Náhodně zvolit, zda počítat Le nebo Lr
 - Když Le:
return weight * Le;
 - Když Lr:
weight = weight * reflectance;
Zvolit nový paprsek $\sim$ BRDF
jít na bod 2.



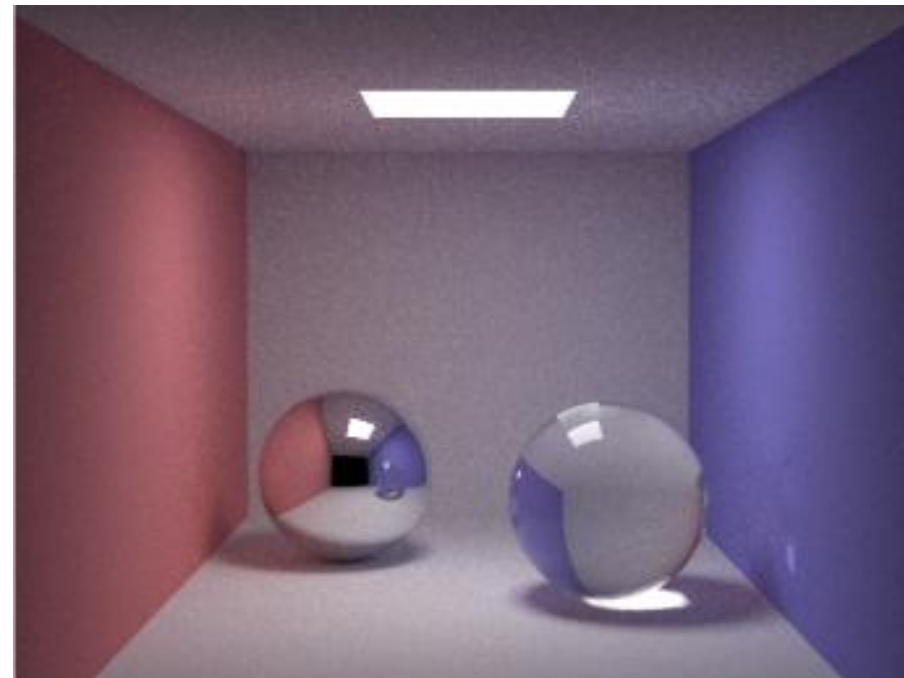
4 paprsky na pixel
16 stínových paprsků na primární p.



64 paprsky na pixel
1 stínový paprsek na primární p.



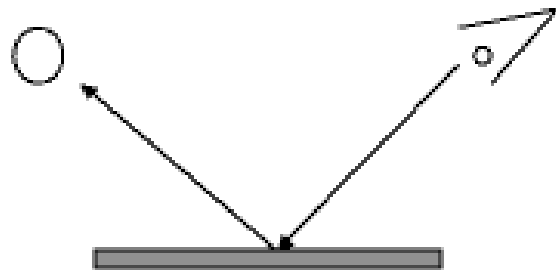
10 paprsků na pixel



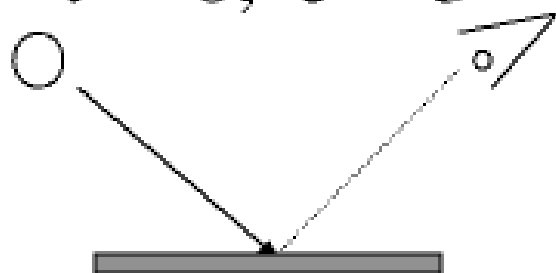
100 paprsků na pixel

1. Zvolit paprsek
 - Vybrat zdroj světla podle funkce rozložení energie
2. Sledovat paprsek
3. Náhodně zvolit, zda absorbovat nebo odrazit (podle reflectance)
 - Když je odražen:
náhodně vygenerovat pokračovatele $\sim$ BRDF
jít na bod 2.
 - Když je pohlcen v kameře:
zaznamenat weight v bodě (x,y)
jít na bod 1.

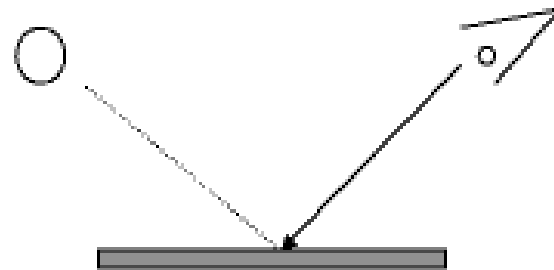
$$k = l + e$$



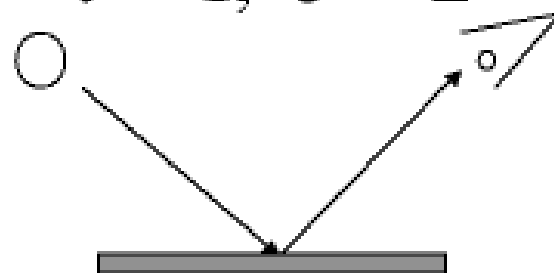
$$l = 0, e = 3$$



$$l = 2, e = 1$$



$$l = 1, e = 2$$



$$l = 3, e = 0$$

$$k = 3$$



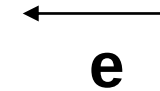
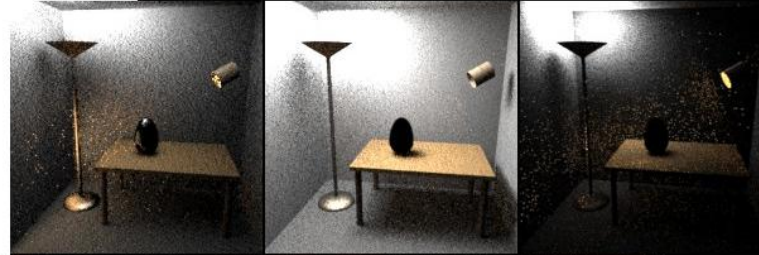
- Jako Distributed ray tracing
 - Cesta z oka, podle distribuční funkce se odráží
 - Při odrazu možno vypočítat stínové paprsky
 - Na konci vypočítat stínový paprsek
- Opačný směr
 - Cesta ze světelného zdroje
 - Na konci „stínový“ paprsek ke kameře
 - Photon tracing, Photon maps
- Dvousměrně
 - Cesta z oka
 - Cesta z kamery
 - Spojit

$k=3$

$(l=2, e=1)$

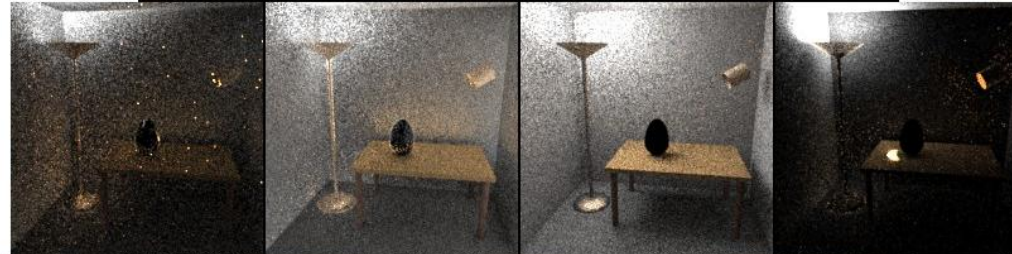


$k=4$



e

$k=5$



$k=6$



$(l=5, e=1)$

$k=7$

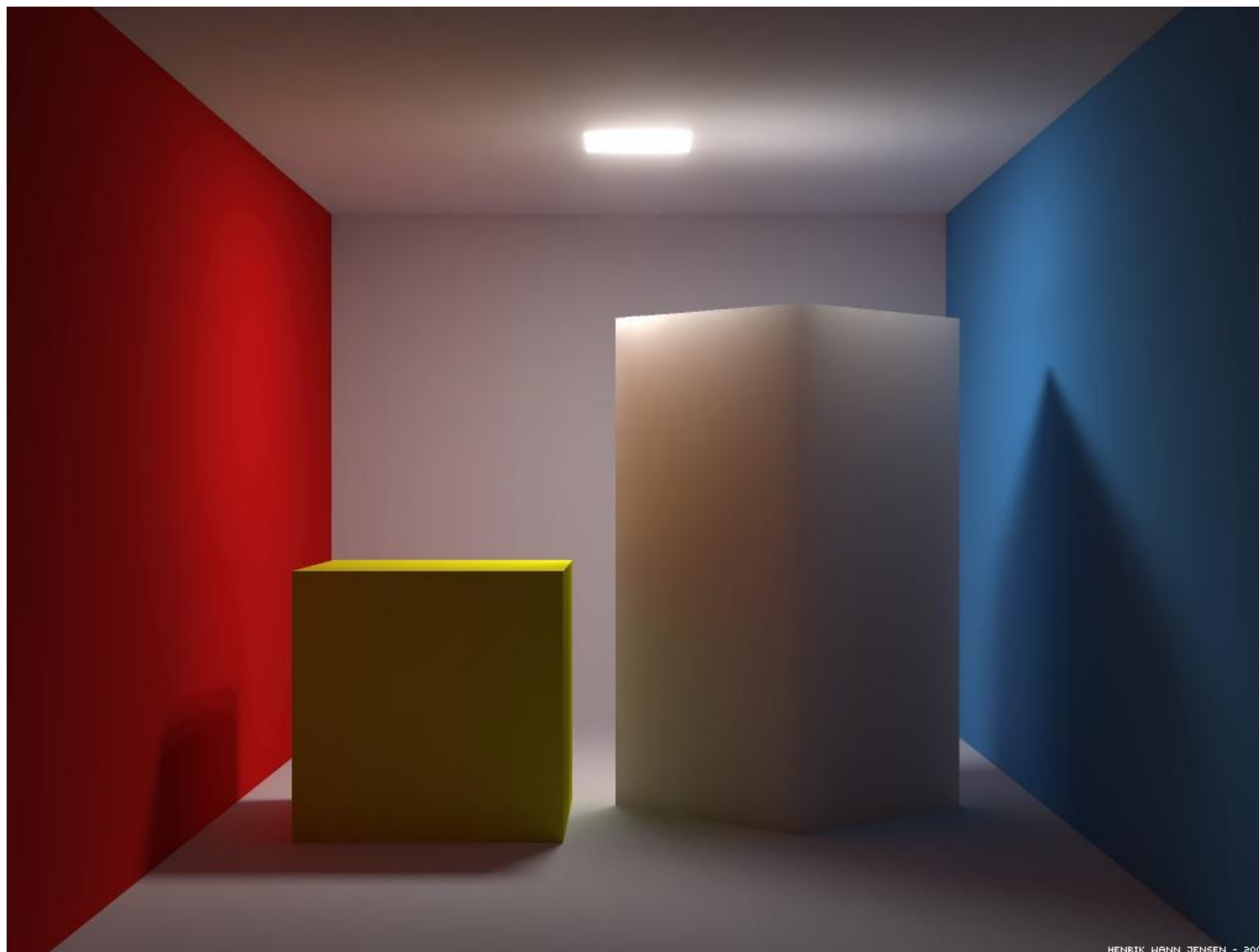




Bidirectional Path Tracing



Path Tracing (Distributed Ray Tracing)



HENRIK WANN JENSEN - 2002

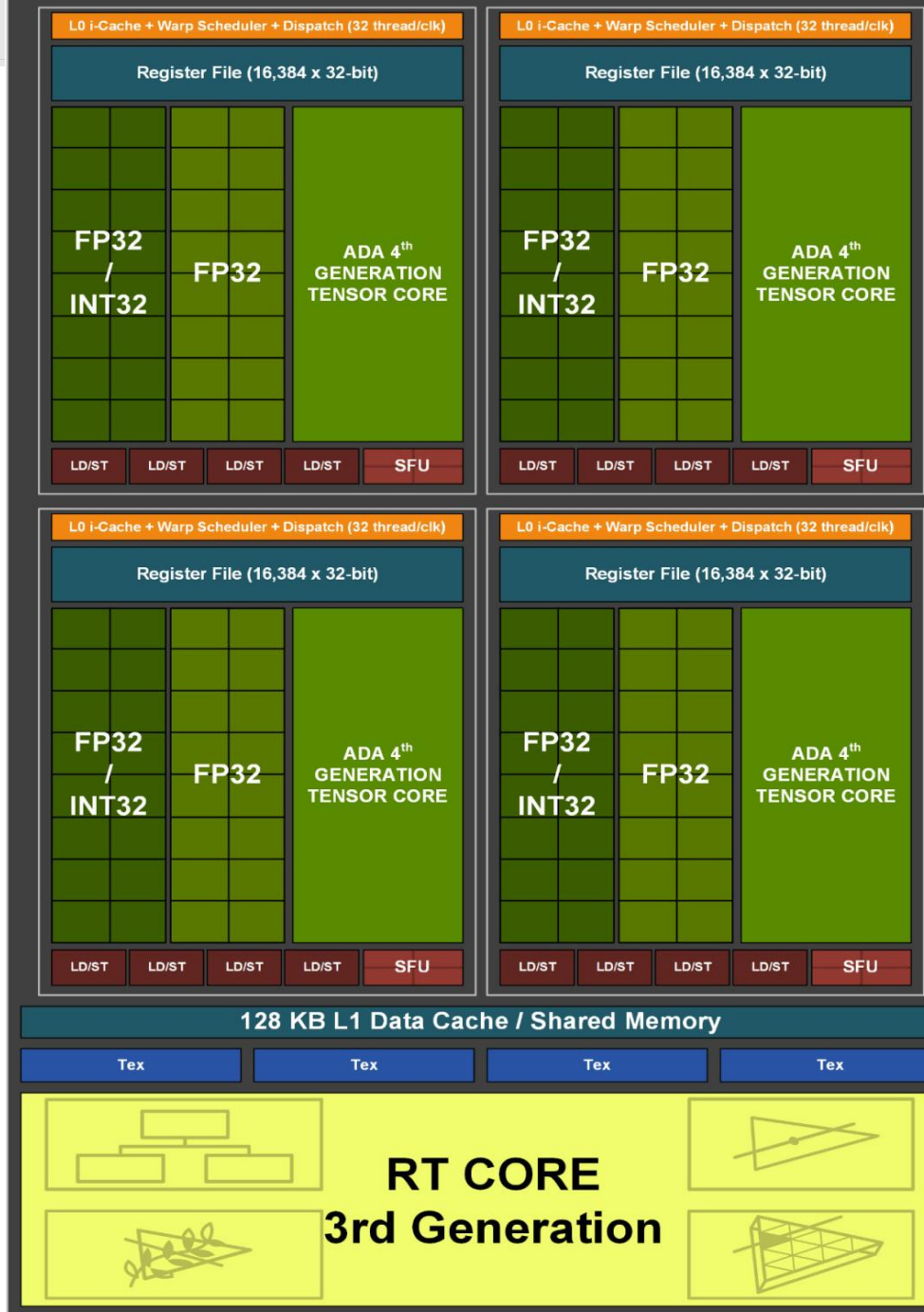
- Light maps
 - Multitexturing v OpenGL
- Environment lighting

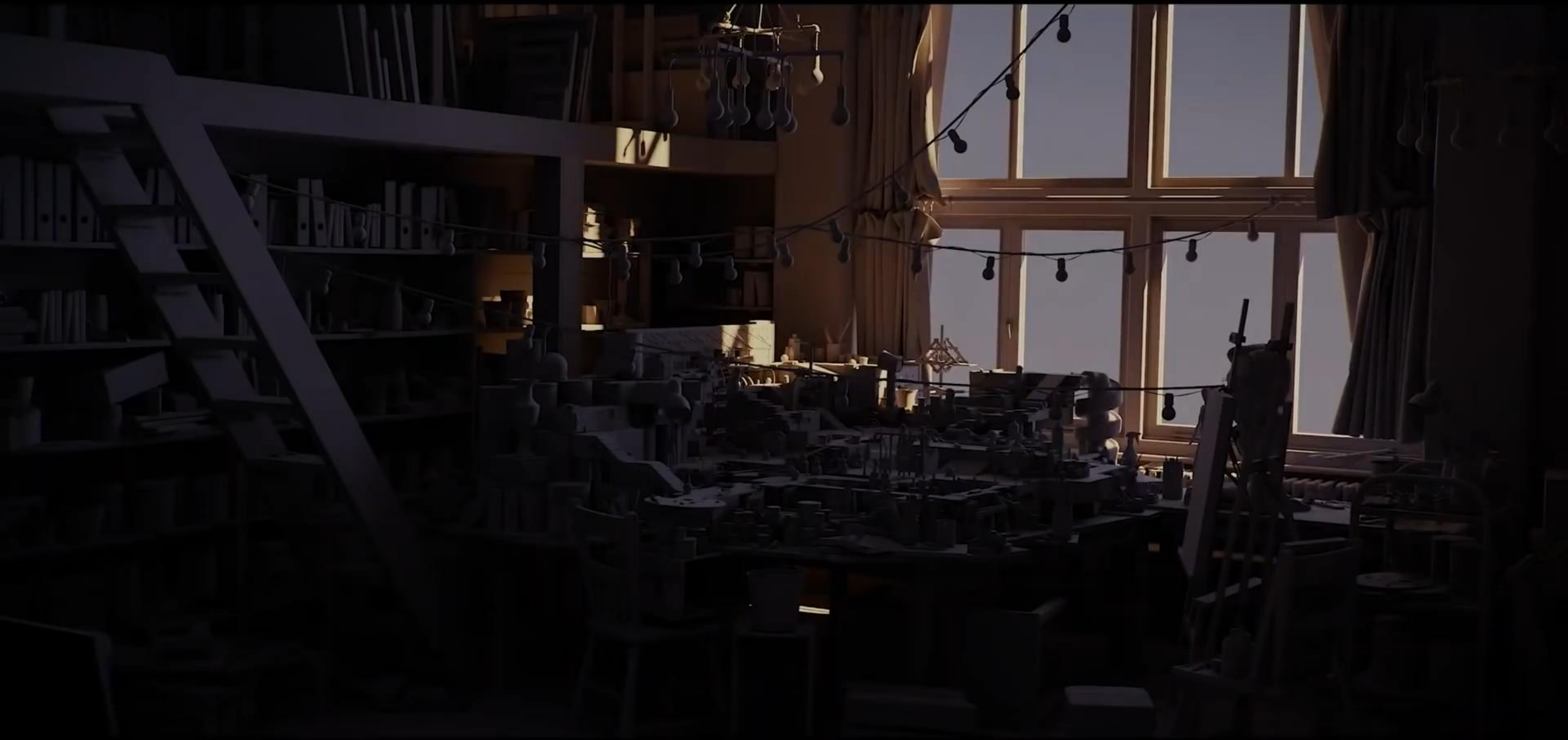


Triple Product Wavelet Integrals For All-Frequency Relighting

Ren Ng	Stanford University
Ravi Ramamoorthi	Columbia University
Pat Hanrahan	Stanford University

SM





- Distributed ray tracing
 - Paprsek (primární či sekundární) není určen přesně, ale distribuční funkcí
- Radiozita
 - Fyzikálně správné osvětlení pro DIFÚZNÍ povrchy
- Photon Mapping
- Path tracing, Bidirectional Path Tracing